

HSPT Exam Simulator

Full-Length Practice Exam 05

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Which word is closest in meaning to meticulous?

- A) ordinary
- B) speedy
- C) precise
- D) careless

2. Which word is closest in meaning to alleviate?

- A) predict
- B) collect
- C) relieve
- D) announce

3. Which word is closest in meaning to candid?

- A) uncertain
- B) silent
- C) honest
- D) formal

4. Which word is closest in meaning to resilient?

- A) fragile
- B) adaptable
- C) distant
- D) decorative

5. Which word is closest in meaning to concise?

- A) brief
- B) confusing
- C) ancient
- D) cheerful

6. Which word is closest in meaning to obsolete?

- A) portable
- B) unfinished
- C) valuable
- D) outdated

7. Which word is closest in meaning to fervent?

- A) casual
- B) doubtful
- C) patient
- D) passionate

8. Which word is closest in meaning to impartial?

- A) experienced
- B) strict
- C) fair
- D) helpful

9. Which word is closest in meaning to precarious?

- A) uncertain
- B) balanced
- C) remote
- D) valuable

10. Which word is closest in meaning to augment?

- A) replace
- B) increase
- C) observe
- D) separate

11. Which word is closest in meaning to tranquil?

- A) shallow
- B) peaceful
- C) brilliant
- D) crowded

12. Which word is closest in meaning to discern?

- A) scatter
- B) forgive
- C) construct
- D) recognize

13. Which word is most nearly opposite in meaning to abundant?

- A) useful
- B) scarce
- C) fresh
- D) hidden

14. Which word is most nearly opposite in meaning to rigid?

- A) narrow
- B) flexible
- C) smooth
- D) heavy

15. Which word is most nearly opposite in meaning to transparent?

- A) colorless
- B) polished
- C) opaque
- D) fragile

16. Which word is most nearly opposite in meaning to reluctant?

- A) careful
- B) unable
- C) quiet
- D) eager

17. Which word is most nearly opposite in meaning to novice?

- A) leader
- B) visitor
- C) student
- D) expert

18. Which word is most nearly opposite in meaning to expand?

- A) arrange
- B) measure
- C) repair
- D) contract

19. Which word is most nearly opposite in meaning to permanent?

- A) complete
- B) reliable
- C) temporary
- D) visible

20. Which word is most nearly opposite in meaning to courteous?

- A) rude
- B) formal
- C) timid
- D) serious

21. Which word is most nearly opposite in meaning to artificial?

- A) colorful
- B) complex
- C) natural
- D) rare

22. Which word is most nearly opposite in meaning to optimistic?

- A) pessimistic
- B) energetic
- C) curious
- D) practical

23. Which word is most nearly opposite in meaning to vague?

- A) reasonable
- B) specific
- C) familiar
- D) brief

24. Which word is most nearly opposite in meaning to hostile?

- A) distant
- B) cautious
- C) powerful
- D) friendly

25. Seismograph is to earthquakes as stethoscope is to ____.

- A) vision
- B) medicine
- C) temperature
- D) heartbeats

26. Blueprint is to building as recipe is to ____.

- A) kitchen
- B) chef
- C) plate
- D) meal

27. Odometer is to distance as scale is to ____.

- A) time
- B) height
- C) weight
- D) volume

28. Cub is to bear as eaglet is to ____.

- A) nest
- B) flock
- C) eagle
- D) feather

29. Gavel is to judge as baton is to ____.

- A) conductor
- B) speaker
- C) musician
- D) athlete

30. Petal is to flower as scale is to ____.

- A) pond
- B) fish
- C) fin
- D) shell

31. Whisper is to quiet as glare is to ____.

- A) sudden
- B) warm
- C) bright
- D) angry

32. Architect is to design as translator is to ____.

- A) interpret
- B) travel
- C) publish
- D) memorize

33. Island is to archipelago as star is to ____.

- A) galaxy
- B) constellation
- C) orbit
- D) planet

34. Sap is to tree as blood is to ____.

- A) vein
- B) oxygen
- C) heart
- D) animal

35. Cautious is to reckless as generous is to ____.

- A) grateful
- B) stingy
- C) kind
- D) wealthy

36. Chapter is to novel as movement is to ____.

- A) symphony
- B) concert
- C) instrument
- D) melody

37. Which word does not belong with the others?

- A) limestone
- B) marble
- C) copper
- D) basalt

38. Elm, birch, and maple belong to the same group. Which word belongs with them?

- A) ivy
- B) oak
- C) fern
- D) moss

39. Which word does not belong with the others?

- A) rectangle
- B) pentagon
- C) triangle
- D) sphere

40. Violin, cello, and viola belong to the same group. Which word belongs with them?

- A) clarinet
- B) trumpet
- C) harp
- D) trombone

41. Which word does not belong with the others?

- A) applaud
- B) mumble
- C) whisper
- D) murmur

42. January, March, and May belong to the same group. Which month belongs with them?

- A) June
- B) July
- C) September
- D) April

43. Which word does not belong with the others?

- A) turquoise
- B) crimson
- C) maroon
- D) scarlet

44. Biography, memoir, and autobiography belong to the same group. Which word belongs with them?

- A) sonnet
- B) fable
- C) journal
- D) drama

45. Which word does not belong with the others?

- A) erode
- B) freeze
- C) condense
- D) evaporate

46. Carpenter, mason, and electrician belong to the same group. Which word belongs with them?

- A) plumber
- B) merchant
- C) pilot
- D) editor

47. Which word does not belong with the others?

- A) venus
- B) mercury
- C) saturn
- D) sirius

48. Meter, liter, and gram belong to the same group. Which word belongs with them?

- A) fraction
- B) second
- C) triangle
- D) balance

49. All cedar boxes are wooden. No wooden object is transparent. Which conclusion must be true?

- A) No cedar box is transparent.
- B) All boxes are wooden.
- C) Every wooden object is a cedar box.
- D) Some transparent objects are boxes.

50. Four runners—Jade, Luis, Mina, and Owen—finish a race. Jade finishes before Mina. Owen finishes after Mina but before Luis. Who must finish first?

- A) Luis
- B) Mina
- C) Jade
- D) Owen

51. A museum tour is offered only on Wednesday or Saturday. Caleb cannot attend Saturday. If Caleb takes the tour, which statement must be true?

- A) Caleb attends Wednesday.
- B) Caleb visits twice.
- C) The tour is offered only Wednesday.
- D) The Saturday tour is canceled.

52. If the library is open, then the front lights are on. The front lights are not on. What must be true?

- A) The library is not open.
- B) The library opens later.
- C) A librarian is absent.
- D) The lights are broken.

53. Nora is taller than Eli. Eli is taller than Priya. Sam is shorter than Priya. Who is tallest?

- A) Priya
- B) Sam
- C) Eli
- D) Nora

54. Every member of the debate team attends practice. Talia is a member of the debate team. Which conclusion follows?

- A) Talia leads the debate team.
- B) Talia attends practice.
- C) Only team members attend practice.
- D) Practice occurs daily.

55. A password contains exactly one letter followed by two different digits. The letter is R, and the first digit is 4. Which could be the password?

- A) 47R
- B) RR4
- C) R47
- D) R44

56. Ben, Ana, and Cole each choose a different club: art, chess, or robotics. Ana does not choose art. Cole chooses chess. Ben does not choose robotics. Which club does Ana choose?

- A) robotics
- B) It cannot be determined.
- C) chess
- D) art

57. A red card is somewhere to the left of a blue card. A green card is immediately right of the red card. Which order could be correct?

- A) green, red, blue
- B) blue, red, green
- C) red, blue, green
- D) red, green, blue

58. No reptiles have feathers. All lizards are reptiles. Which statement must be true?

- A) All reptiles are lizards.
- B) Some feathered animals are lizards.
- C) No reptiles have scales.
- D) No lizards have feathers.

59. A train stops at Pine before Oak and at Oak before River. It stops at Lake after River. At which station does it stop second?

- A) Lake
- B) Oak
- C) Pine
- D) River

60. If a shape is a square, then it has four equal sides. Shape X does not have four equal sides. Which conclusion is valid?

- A) Shape X has no equal sides.
- B) Shape X is not a square.
- C) Every four-sided shape is a square.
- D) Shape X is a triangle.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. What number comes next? 4, 9, 19, 39, 79, ____

- A) 157
- B) 159
- C) 161
- D) 119

62. What number comes next? 96, 82, 69, 57, 46, ____

- A) 34
- B) 35
- C) 36
- D) 37

63. What number comes next? 3, 6, 5, 10, 9, 18, 17, ____

- A) 34
- B) 33
- C) 35
- D) 36

64. What number comes next? 5, 8, 14, 23, 35, ____

- A) 48
- B) 49
- C) 47
- D) 50

65. What number comes next? 6, 12, 20, 30, 42, ____

- A) 56
- B) 52
- C) 54
- D) 48

66. What number comes next? 243, 81, 27, 9, ____

- A) 2
- B) 6
- C) 3
- D) 1

67. What number comes next? 7, 10, 16, 25, 37, ____

- A) 49
- B) 51
- C) 50
- D) 52

68. What number comes next? 1, 4, 10, 22, 46, ____

- A) 96
- B) 90
- C) 94
- D) 92

69. What number comes next? 64, 32, 16, 8, ____

- A) 6
- B) 4
- C) 2
- D) 7

70. What number comes next? 0.2, 0.6, 1.8, 5.4, ____

- A) 16.2
- B) 21.6
- C) 15.2
- D) 10.8

71. What number comes next? 11, 15, 23, 39, 71, ____

- A) 119
- B) 103
- C) 135
- D) 127

72. What number comes next? 2, 3, 5, 8, 12, 17, ____

- A) 24
- B) 21
- C) 23
- D) 22

73. What number comes next? 100, 97, 91, 82, 70, ____

- A) 54
- B) 55
- C) 52
- D) 58

74. What number comes next? 4, 7, 13, 25, 49, ____

- A) 97
- B) 95
- C) 96
- D) 98

75. A number is multiplied by 4, and then 9 is added. The result is 45. What is the original number?

- A) 7
- B) 11
- C) 8
- D) 9

76. What is 37.5% of 64?

- A) 22
- B) 26
- C) 20
- D) 24

77. The ratio of blue beads to white beads is 5:3. If there are 40 blue beads, how many white beads are there?

- A) 24
- B) 30
- C) 32
- D) 18

78. Which value is equivalent to $2\frac{3}{5} \div 1\frac{3}{10}$?

- A) 2
- B) 1
- C) $2\frac{1}{2}$
- D) 3

79. The average of 18, 23, 27, and n is 24. What is n ?

- A) 26
- B) 27
- C) 29
- D) 28

80. A jacket priced at \$72 is discounted by 25%. What is the sale price?

- A) \$18
- B) \$54
- C) \$48
- D) \$60

81. If 7 notebooks cost \$24.50, what is the cost of 3 notebooks at the same rate?

- A) \$10.50
- B) \$9.50
- C) \$10.00
- D) \$11.50

82. Which expression has a value of 17?

- A) $2(9) + 1$
- B) $3(6) - 1$
- C) $4(5) - 2$
- D) $5(4) - 1$

83. A tank is $\frac{3}{5}$ full. After 18 liters are added, it is $\frac{9}{10}$ full. What is the tank's capacity?

- A) 72 liters
- B) 54 liters
- C) 45 liters
- D) 60 liters

84. What is the least positive integer divisible by 9, 12, and 15?

- A) 180
- B) 360
- C) 90
- D) 120

85. A map uses a scale of 1 inch to 18 miles. Two towns are 3.5 inches apart on the map. What is the actual distance?

- A) 54 miles
- B) 72 miles
- C) 63 miles
- D) 60 miles

86. When 8 is added to twice a number, the result is 50. What is the number?

- A) 18
- B) 20
- C) 21
- D) 29

87. A recipe for 6 servings uses $2\frac{1}{4}$ cups of rice. How many cups are needed for 10 servings?

- A) 3
- B) $3\frac{3}{4}$
- C) $3\frac{1}{2}$
- D) 4

88. What is the value of $|-18| - \sqrt{81}$?

- A) 9
- B) 27
- C) 7
- D) 18

89. Quantity A: $\frac{7}{12}$ of 60

Quantity B: 35

- A) The relationship cannot be determined from the information given.
- B) Quantity A is greater.
- C) Quantity B is greater.
- D) The two quantities are equal.

90. Quantity A: 45% of 80

Quantity B: $\frac{3}{8}$ of 88

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

91. Quantity A: x

Quantity B: 11

Given that $5x - 7 = 48$.

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) Quantity A is greater.

92. Quantity A: The greatest common factor of 42 and 70

Quantity B: 12

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

93. Quantity A: The number of minutes in 2.5 hours

Quantity B: The number of seconds in 2.5 minutes

- A) The relationship cannot be determined from the information given.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) Quantity A is greater.

94. Quantity A: 0.72×50

Quantity B: $3^2 \times 4$

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

95. Quantity A: The remainder when 73 is divided by 8

Quantity B: The remainder when 59 is divided by 6

- A) Quantity B is greater.
- B) The two quantities are equal.
- C) The relationship cannot be determined from the information given.
- D) Quantity A is greater.

96. Quantity A: The median of 4, 9, 12, 18, 25

Quantity B: The mean of 8, 12, and 16

- A) Quantity B is greater.
- B) Quantity A is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

97. Quantity A: $|-23|$

Quantity B: $\sqrt{529}$

- A) Quantity B is greater.
- B) The relationship cannot be determined from the information given.
- C) Quantity A is greater.
- D) The two quantities are equal.

98. Quantity A: 0.625

Quantity B: $\frac{5}{8}$

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) Quantity A is greater.

99. Quantity A: y

Quantity B: 15

Given that $\frac{y}{3} + 4 = 10$.

- A) The two quantities are equal.
- B) The relationship cannot be determined from the information given.
- C) Quantity B is greater.
- D) Quantity A is greater.

100. Quantity A: The number of positive factors of 24

Quantity B: 7

- A) Quantity B is greater.
- B) The relationship cannot be determined from the information given.
- C) Quantity A is greater.
- D) The two quantities are equal.

101. Quantity A: The area of a semicircle with radius 6 cm, using $\pi \approx 3.14$

Quantity B: The area of a triangle with base 14 cm and height 8 cm

- A) The two quantities are equal.
- B) The relationship cannot be determined from the information given.
- C) Quantity B is greater.
- D) Quantity A is greater.

102. Quantity A: The perimeter of an equilateral triangle with side length 12 cm

Quantity B: The perimeter of a rectangle 11 cm long and 7 cm wide

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The two quantities are equal.

103. Quantity A: The volume of a cube with edge length 5 cm

Quantity B: 120 cm^3

- A) The two quantities are equal.
- B) The relationship cannot be determined from the information given.
- C) Quantity A is greater.
- D) Quantity B is greater.

104. Quantity A: The diameter of a circle with radius 13 cm

Quantity B: 25 cm

- A) Quantity B is greater.
- B) Quantity A is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

105. Quantity A: The third angle of a triangle whose other angles are 52° and 71°
Quantity B: 57°

- A) The relationship cannot be determined from the information given.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) Quantity A is greater.

106. Quantity A: The area of a triangle with base 15 cm and height 8 cm
Quantity B: The area of a parallelogram with base 10 cm and height 6 cm

- A) The two quantities are equal.
- B) Quantity A is greater.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

107. Quantity A: The circumference of a circle with radius 4 cm, using $\pi \approx 3.14$
Quantity B: 25 cm

- A) Quantity B is greater.
- B) Quantity A is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

108. Quantity A: The perimeter of a regular hexagon with side length 6 cm
Quantity B: The perimeter of a square with side length 9 cm

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) Quantity A is greater.

109. Quantity A: The volume of a rectangular prism measuring 3 cm by 5 cm by 7 cm
Quantity B: 100 cm^3

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) Quantity B is greater.
- D) The two quantities are equal.

110. Quantity A: The area of a circle with radius 3 cm
Quantity B: 30 cm^2
Use $\pi \approx 3.14$.

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

111. Quantity A: The number of diagonals in a rectangle
Quantity B: The number of diagonals in a pentagon

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

112. Quantity A: The sum of the interior angles of a quadrilateral
Quantity B: 360°

- A) The relationship cannot be determined from the information given.
- B) Quantity A is greater.
- C) Quantity B is greater.
- D) The two quantities are equal.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: The First Clear Image

When Leila arrived at the neighborhood observatory, she expected the telescope to deliver an impressive view as soon as she looked through it. Instead, Jupiter appeared as a pale blur. She turned the focus knob quickly, first one way and then the other, but the image only swelled and shrank.

Mr. Alvarez, the volunteer beside her, did not reach for the controls. He asked Leila to point the smaller finder scope at a bright rooftop antenna. Through the finder, the antenna sat near the edge of the crosshairs rather than at their center. Leila adjusted three tiny screws until the crosshairs rested precisely on it. Then she aimed the main telescope at the same antenna and centered that view as well.

"The two scopes have to agree about where they are looking," Mr. Alvarez explained. "Otherwise, finding a planet is mostly guesswork."

By the time darkness deepened, Leila had repeated the alignment twice. Her first attempt at Jupiter still looked soft, but now she resisted the urge to spin the knob. She moved it slowly and paused after each slight turn. A sharp disk emerged, crossed by two brownish bands. Four bright points appeared in a line nearby.

Leila stepped away so the next visitor could look. The image had lasted only a moment before the atmosphere made it tremble, yet she no longer felt disappointed. What pleased her most was not simply seeing Jupiter. It was knowing why the view had become clear.

113. What was wrong with the finder scope when Leila first checked it?

- A) Its adjustment screws were missing.
- B) The antenna appeared near the edge of the crosshairs.
- C) Its lens was covered with dust.
- D) It pointed at Jupiter instead of the antenna.

114. Why does Mr. Alvarez initially avoid adjusting the telescope himself?

- A) He is waiting for the atmosphere to clear.
- B) He believes the telescope is damaged.
- C) He wants Leila to learn the alignment process.
- D) He does not know how the controls work.

115. As used in the final paragraph, "tremble" most nearly means

- A) shake slightly.
- B) vanish completely.
- C) change color.
- D) grow brighter.

116. Which statement best expresses a central idea of the passage?

- A) Atmospheric conditions make planet viewing impossible.
- B) Understanding a process can be as rewarding as obtaining the result.
- C) Observatories should allow only trained adults to use telescopes.
- D) Scientific equipment works best without human adjustment.

117. What does Leila's slower movement of the focus knob show about her?

- A) She is afraid the telescope will break.
- B) She no longer cares whether Jupiter is visible.
- C) She has become more patient and deliberate.
- D) She has forgotten Mr. Alvarez's directions.

Informational Passage: Farmers Beneath the Forest Floor

Leafcutter ants do not eat most of the leaf pieces they carry underground. Instead, they use the plant material to cultivate a particular fungus, which serves as their main food. In this way, a colony operates something like a farm hidden beneath the forest floor.

Workers cut leaves into manageable sections and transport them to the nest. Other ants clean the pieces, chew them into a soft pulp, and add them to carefully maintained fungus gardens. The ants remove unwanted organisms and carry waste to separate chambers. Even temperature and moisture matter: tunnels and openings help air circulate through the nest.

The partnership benefits both species. The fungus receives a steady supply of prepared plant material and a protected place to grow. The ants receive nutritious structures produced by the fungus. Neither partner involved in this close relationship commonly survives in the same way without the other.

Leafcutter colonies can remove large amounts of vegetation, so farmers sometimes view them only as pests. Ecologists, however, also study how the ants affect soil. By moving organic material underground and excavating tunnels, the colonies redistribute nutrients and change how air and water travel through the ground. Their farming may damage a garden, but it also demonstrates that agriculture is not an exclusively human invention.

118. What is the ants' main food according to the passage?

- A) seeds stored in waste chambers
- B) structures produced by cultivated fungus
- C) the fresh leaves carried into the nest
- D) other insects that enter the tunnels

119. Why are tunnels and openings important to the fungus gardens?

- A) They allow sunlight to reach the fungus directly.
- B) They prevent ants from carrying in leaves.
- C) They help regulate conditions by circulating air.
- D) They separate young ants from workers.

120. As used in the final paragraph, "redistribute" most nearly means

- A) measure accurately.
- B) move into different locations.
- C) hide from view.
- D) destroy completely.

121. What is the main purpose of the passage?

- A) To describe how farmers can remove underground nests
- B) To prove that all ants cause serious agricultural damage
- C) To explain how leafcutter ants cultivate fungus and affect their environment
- D) To compare the sizes of several forest insects

122. How does the final paragraph develop the discussion?

- A) It argues that fungi are harmful to every forest.
- B) It repeats the exact steps used to cut and clean leaves.
- C) It broadens the focus from the colony's farm to its environmental effects.
- D) It presents a chronological history of ant research.

Humanities Passage: Geometry Stitched in Cloth

A quilt may first attract attention through color, but its construction also depends on measurement and geometry. Quilters divide cloth into shapes, arrange those shapes into repeating units called blocks, and join the blocks to form a larger design. A small error in one piece can multiply across an entire quilt.

Many traditional patterns transform simple figures through repetition. Two triangles can form a square; several squares can produce a checkerboard; rotated diamonds can create the impression of stars. Symmetry often gives a pattern visual balance, while a deliberate break in symmetry can draw the eye toward one area.

The mathematics is practical as well as decorative. Before cutting fabric, a quilter must account for seam allowances—the narrow edges that disappear inside each seam. A visible square that should measure four inches across must begin slightly larger. Quilters also estimate how many blocks will cover a bed and how borders will affect the final dimensions.

Digital design programs now allow artists to test colors and arrangements before cutting any cloth. Yet many quilters still sketch on graph paper or place fabric pieces on a wall. These methods make it easy to step back, notice an imbalance, and rearrange the design.

Recognizing the geometry in quilts does not reduce them to exercises in calculation. Instead, it reveals one source of their artistry. Accurate measurement creates the structure within which color, memory, and invention can take shape.

123. What are quilt blocks?

- A) pieces used only along a quilt's border
- B) digital copies of traditional patterns
- C) narrow edges hidden inside seams
- D) repeating units joined into a larger design

124. Why must a four-inch visible square begin slightly larger than four inches?

- A) The border will cover the square completely.
- B) Rotating the square increases its width.
- C) The cloth always shrinks before it is sewn.
- D) Part of the fabric will be taken up by seam allowances.

125. Why might a quilter place pieces on a wall before sewing them?

- A) To calculate the price of the finished quilt
- B) To stretch every piece to the same size
- C) To evaluate the overall balance from a distance
- D) To prevent the fabric from losing color

126. As used in the second paragraph, “deliberate” most nearly means

- A) intentional.
- B) traditional.
- C) uncertain.
- D) slow.

127. Which statement best expresses the passage's main idea?

- A) Mathematical structure contributes to the practical construction and artistry of quilts.
- B) Quilters value measurement more than color or memory.
- C) Digital programs have made traditional quilting unnecessary.
- D) Every successful quilt must use perfect symmetry.

128. What is the effect of the phrase “the structure within which color, memory, and invention can take shape”?

- A) It argues that memories can be measured mathematically.
- B) It suggests that quilts should be judged only by their dimensions.
- C) It emphasizes that precision supports rather than limits creativity.
- D) It shows that modern tools produce less creative designs.

Literary Passage: A Bench for the Courtyard

The cedar boards looked perfectly straight when Malik selected them, but on the worktable one corner of the bench frame refused to lie flat. He pressed down. The opposite corner lifted like the lid of a box.

“The floor might be uneven,” he suggested.

Ms. Patel slid the frame to another table. It rocked there too. She handed Malik a measuring tape and a carpenter's square, then returned to helping another student.

Malik measured every board again. Each length matched the plan. Only when he checked the diagonals did he find the problem: one diagonal was half an inch longer than the other. The frame was not rectangular but slightly skewed.

He loosened the corner clamps, nudged the boards, and measured both diagonals repeatedly. When the numbers finally matched, he tightened the clamps and tested the frame. It stayed level.

The next afternoon, students carried the finished bench into the courtyard. Malik noticed no one examining its corners or praising its equal diagonals. They simply sat down, set their backpacks beside them, and continued talking. For a moment he wished the hidden correction were more visible. Then he watched the bench support three students without wobbling. The successful measurement, he realized, was visible after all—just not as a number.

129. How does Malik determine that the frame is skewed?

- A) One board is shorter than the plan requires.
- B) The two diagonal measurements are different.
- C) The carpenter's square is damaged.
- D) The frame rocks only on the first table.

130. Why does Ms. Patel give Malik tools instead of identifying the problem for him?

- A) She has not noticed that the frame rocks.
- B) She wants him to investigate the cause himself.
- C) She thinks the bench plan is incorrect.
- D) She expects another student to repair it.

131. As used in the fifth paragraph, “nudged” most nearly means

- A) measured twice.
- B) struck forcefully.
- C) replaced completely.
- D) moved slightly.

132. How is the passage mainly organized?

- A) Several bench designs are compared from least to most expensive.
- B) A disagreement is presented and left unresolved.
- C) Instructions are listed for selecting and cutting cedar boards.
- D) A construction problem is discovered, investigated, corrected, and reconsidered.

133. What does Malik mean when he decides the correction is visible “just not as a number”?

- A) The bench looks larger than the original plan.
- B) The students can identify every change he made.
- C) The diagonal lengths have been written beneath the bench.
- D) The stable bench demonstrates the value of the accurate measurement.

Informational Passage: Life Between the Tides

A tide pool may seem calm when the ocean withdraws, but its inhabitants face rapid changes. During high tide, waves cover the pool with cool seawater. Hours later, sunlight may warm the trapped water while evaporation makes it saltier. Rain can create the opposite effect by diluting the pool.

Organisms survive these shifts in different ways. Anemones close their bodies to reduce water loss. Crabs hide beneath stones, where temperatures change more slowly. Some snails seal moisture inside their shells.

Seaweeds bend under waves instead of resisting them, and strong attachments keep them from being swept away.

Location matters. Pools high on the shore remain separated from the ocean longer than lower pools, so their temperatures and salt levels often vary more. Lower pools are refreshed more frequently but may experience stronger wave action and more visits from fish searching for food.

Scientists studying tide pools record more than which species are present. They may measure water temperature, salinity, depth, and the length of time a pool remains exposed. Comparing these observations across many tides helps explain why a species thrives in one part of the shore but not another. A tide pool is therefore not merely a small container of seawater. It is a changing habitat shaped by both land and ocean.

134. Which organism closes its body to reduce water loss?

- A) an anemone
- B) a crab
- C) a snail
- D) a seaweed

135. Why do high-shore pools often experience wider changes in salinity?

- A) They are always deeper than lower pools.
- B) They receive stronger waves during every tide.
- C) They remain cut off from fresh seawater for longer periods.
- D) They contain no organisms that use salt.

136. Which statement best expresses the main idea?

- A) Tide pools are variable habitats whose organisms use different strategies to survive.
- B) Rain is the greatest danger to every organism in a tide pool.
- C) Tide-pool species depend mainly on protection from fish.
- D) Only low-shore tide pools support a wide range of life.

137. Why does the author describe a tide pool as a “changing habitat shaped by both land and ocean”?

- A) To suggest that every tide pool moves to a new location
- B) To show that land organisms dominate ocean organisms
- C) To prove that tide pools are artificial containers
- D) To emphasize that conditions vary through interacting influences

138. What is the effect of the opening contrast between a pool that “may seem calm” and the changes its inhabitants face?

- A) It suggests that tide-pool organisms are inactive.
- B) It creates an argument between two researchers.
- C) It shows that the author doubts scientific measurements.
- D) It challenges a first impression and introduces hidden difficulty.

Social Science Passage: What Old Trash Can Tell Us

Archaeologists do not study only temples, jewelry, and carefully made tools. They also examine discarded fish bones, broken pottery, worn-out shoes, and other materials that earlier communities considered useless. A concentrated deposit of household refuse is sometimes called a midden.

Objects in a midden can reveal ordinary habits that formal records ignore. Animal bones may indicate what people ate and which seasons they occupied a site. Seeds and pollen can provide evidence about local plants. Marks on pottery may show how vessels were made or repaired. Even the location of waste can suggest how a settlement organized space.

Interpreting refuse requires caution. A missing object does not prove that people never used it; materials such as wood or cloth may decay. Later digging, floods, or animals can mix layers that were deposited at different times. Researchers therefore record the exact position of each find and compare several kinds of evidence.

Modern waste studies can also challenge assumptions. People often report that they recycle more or waste less than their discarded materials actually show. The difference does not necessarily mean that individuals intend to be dishonest. Memory is selective, and people may describe what they usually do rather than every exception. Trash is valuable to researchers precisely because it was not usually arranged to impress an audience. The remains are incomplete, but when interpreted carefully, they offer a direct view of repeated choices in daily life.

139. What can animal bones in a midden help researchers determine?

- A) the exact language spoken in the settlement
- B) the names of everyone who lived nearby
- C) the market price of household pottery
- D) what people ate and when they may have occupied a site

140. Why do researchers record the exact position of each find?

- A) Layers can be disturbed or mixed over time.
- B) Only objects found at the center of a site are useful.
- C) Every discarded object must be returned to its owner.
- D) Position determines the monetary value of an object.

141. Why might discarded materials differ from what people report about their habits?

- A) Reports may reflect general intentions while trash records exceptions too.
- B) People cannot remember any of their daily activities.
- C) Modern waste contains no evidence about behavior.
- D) Discarded materials are always placed by strangers.

142. As used in the third paragraph, “decay” most nearly means

- A) move.
- B) shine.
- C) harden.
- D) decompose.

143. What is the author's main purpose?

- A) To provide instructions for organizing household waste
- B) To explain how discarded materials can reveal daily behavior when interpreted carefully
- C) To prove that written records are never accurate
- D) To argue that broken objects are more valuable than complete ones

144. Why does the author discuss both ancient middens and modern waste studies?

- A) To show that archaeologists study only recent sites
- B) To show that discarded materials can reveal behavior in different periods
- C) To claim that ancient and modern communities produced identical waste
- D) To compare two methods of manufacturing pottery

Literary Passage: Thirty Seconds of Silence

Jonah had practiced the introduction to the school radio program until he could say it without looking at the page. At 8:05 on Friday morning, the red studio light came on, and he delivered every word smoothly. Then he pressed the button for the recorded interview.

Nothing happened.

The silent seconds seemed to stretch. Jonah checked the computer screen. The audio file was open, but the play marker had not moved. Through the glass, Ms. Reed pointed to the microphone and held up one finger.

Jonah switched the microphone back on. “While we prepare our interview,” he said, “here is a question for homerooms to consider: What is one change that would make lunchtime less wasteful?” He had planned to ask that question after the interview, not before it.

As he spoke, Ms. Reed restarted the audio program. The file began playing, and Jonah faded down his microphone. His hands were still unsteady, but the silence was gone.

After the broadcast, Jonah apologized for changing the order. Ms. Reed shook her head. “Listeners heard a question and then an interview. They didn’t hear the problem you solved.”

On the following Friday, Jonah brought his carefully marked script into the studio. Beside it he placed a blank card. At the top he had written, “If the plan stops, keep the program moving.”

145. What does Ms. Reed do while Jonah speaks into the microphone?

- A) She leaves the studio to find a script.
- B) She asks homerooms to call the station.
- C) She restarts the audio program.
- D) She records a new interview.

146. Why does Jonah bring a blank card to the next broadcast?

- A) He wants space for a backup idea if the planned sequence fails.
- B) He needs to write down listeners' names.
- C) He plans to cancel the recorded interview.
- D) He intends to replace his marked script completely.

147. As used in the fifth paragraph, “faded down” most nearly means

- A) turned the lights off around.
- B) gradually lowered the sound from.
- C) moved farther away from.
- D) forgot the words for.

148. Which statement best expresses a central idea?

- A) Recorded interviews are less reliable than live broadcasts.
- B) A broadcaster should never change the planned order.
- C) Preparation is valuable, but adaptability is also necessary when problems occur.
- D) Technical problems can be prevented by memorizing a script.

149. What is the effect of the separate sentence “Nothing happened”?

- A) It proves that the studio equipment is permanently broken.
- B) It introduces a new narrator into the passage.
- C) It emphasizes the sudden failure and moment of suspense.
- D) It summarizes the entire broadcast after it ends.

Informational Passage: Lichens: A Partnership on Stone

The pale green or orange patches on a boulder may look like a single plant, but a lichen is a partnership. It usually combines a fungus with an alga or a photosynthetic bacterium. The fungus provides structure and helps retain water, while its partner uses sunlight to produce sugars.

This cooperation allows many lichens to live where neither partner would thrive alone. Some grow on bare rock, tree bark, or exposed soil. They do not have roots that draw nutrients from the surfaces they occupy. Instead, they absorb water and dissolved substances from rain, dust, and the air.

Lichens grow slowly, yet they can influence a landscape over long periods. Acids they release and the expansion of their tissues can help break down rock surfaces. Dust and tiny fragments become trapped around them. These materials may contribute to the earliest thin soil in which mosses and small plants later take hold.

Because lichens absorb substances directly from the atmosphere, some species are sensitive to air pollution. Scientists can compare the kinds and abundance of lichens at different locations as one indicator of air quality. This method does not replace instruments that measure particular pollutants, but it can reveal biological effects over time.

What appears to be a simple stain on stone is therefore both a cooperative organism and a small participant in environmental change.

150. What does the photosynthetic partner provide in a lichen?

- A) instruments for measuring pollutants
- B) sugars made using sunlight
- C) a protective layer of soil
- D) roots that enter the rock

151. Why can lichens help scientists study air quality?

- A) They absorb atmospheric substances and some respond to pollution.
- B) They record daily temperatures in visible rings.
- C) They grow at exactly the same rate in all locations.
- D) They remove every pollutant from the air.

152. As used in the third paragraph, “take hold” most nearly means

- A) absorb all available water.
- B) attach to an instrument.
- C) begin to grow successfully.
- D) stop breaking down rock.

153. Which statement best summarizes the passage?

- A) Lichens are plants whose deep roots split rocks quickly.
- B) Lichens are partnerships that survive in difficult places and contribute to environmental change.
- C) Fungi depend entirely on soil made by mature trees.
- D) Every lichen species responds to pollution in the same way.

154. How does the passage develop from the first paragraph to the final paragraph?

- A) It compares two rocks without explaining the organism on them.
- B) It lists lichen colors before giving directions for removing them.
- C) It presents a debate and then announces which scientist won.
- D) It moves from defining lichens to explaining their roles in landscapes and environmental study.

Humanities Passage: Preserving a Voice

An oral history interview is more than a casual conversation recorded for the future. Its purpose is to preserve a person's account of events and experiences in that person's own voice. The interviewer prepares questions, but the narrator—the person being interviewed—helps determine where the story leads.

Open-ended questions often produce the richest responses. Asking “What do you remember about your first day at the factory?” invites a fuller account than asking “Were you nervous?” Follow-up questions can clarify names, dates, or unfamiliar practices without forcing the narrator toward a particular answer.

Memory, however, is not a perfect recording. People may combine events, forget details, or interpret the past through later experience. This does not make oral history useless. It means researchers should distinguish between what an account can show about a person's perspective and what must be checked against other sources.

Preservation also involves choices after the interview. Archivists may create a written transcript, identify speakers, add searchable descriptions, and store copies in more than one location. They must also respect agreements about whether sensitive portions may be shared.

The result is not simply a collection of facts. Tone, pauses, laughter, and emphasis can communicate meaning that a summary would lose. By preserving both information and voice, oral histories add personal dimensions to the study of the past.

155. Why are open-ended questions useful in oral history interviews?

- A) They prevent narrators from choosing the topic.
- B) They guarantee that every date will be correct.
- C) They require only yes-or-no responses.
- D) They invite narrators to give fuller accounts.

156. Why might a researcher compare an oral history with other sources?

- A) To check factual details while still studying the narrator's perspective
- B) To prove that memory contains no useful information
- C) To replace the narrator's voice with an official summary
- D) To remove all personal experiences from the archive

157. As used in the fourth paragraph, “sensitive” most nearly means

- A) private or potentially harmful if disclosed.
- B) easy to hear on a recording.
- C) written in an unfamiliar language.
- D) unlikely to be remembered.

158. What is the main idea of the passage?

- A) Oral history preserves personal accounts and voices but requires careful questioning and interpretation.
- B) Interviewers should correct a narrator whenever a date is uncertain.
- C) Transcripts preserve every feature of a recorded voice.
- D) Oral history is useful only when no written sources exist.

159. Why does the author list “tone, pauses, laughter, and emphasis”?

- A) To show why all interviews should be performed onstage
- B) To describe mistakes that should be removed from recordings
- C) To illustrate qualities of spoken testimony that a summary may not capture
- D) To argue that facts are unimportant in oral history

Science Passage: A Burr and a Better Fastener

In the 1940s, engineer George de Mestral noticed that burrs clung stubbornly to his clothing and his dog's fur after a walk. Rather than merely pulling them off, he examined them under magnification. Tiny hooks on the burrs had caught in loops formed by fibers and hair.

That observation inspired a reusable fastening system. One strip would contain many small hooks, while another would provide soft loops. Pressed together, the hooks would catch; pulled apart, they would release. Turning the idea into a dependable product took years. The hooks needed to be flexible enough to separate without breaking yet firm enough to hold repeatedly. Manufacturing thousands of uniform hooks also required new techniques.

The eventual fastener became useful in clothing, medical equipment, aircraft, and many other settings. Its development is often cited as an example of biomimicry—design that learns from forms or processes found in nature.

Biomimicry is not simple copying. A burr does not need to pass factory inspections, fit standard dimensions, or work through thousands of uses. Designers must identify the useful principle, then adapt it to human needs and manufacturing limits. De Mestral's achievement began with close observation, but persistence transformed that observation into a practical technology.

160. What feature of burrs inspired the fastening system?

- A) Tiny hooks catch in loops of fiber or hair.
- B) Burrs change shape when warmed.
- C) Their stems produce a strong adhesive.
- D) Smooth seeds slide across clothing.

161. Why did developing the fastener take years after the original observation?

- A) Burrs could be found only during one short season.
- B) No one understood how hooks attach to loops.
- C) The design worked only with animal fur.
- D) A natural principle had to be adapted into a durable, manufacturable product.

162. Which statement best expresses the passage's central idea?

- A) The most useful inventions are created accidentally and immediately.
- B) Factory standards prevent designers from using natural models.
- C) Close observation of nature can inspire technology, but practical design requires further problem solving.
- D) Natural structures can always be copied without modification.

163. How is the passage mainly organized?

- A) It lists uses of burrs in chronological order.
- B) It compares several inventors competing to solve one problem.
- C) It gives step-by-step directions for manufacturing hooks.
- D) It traces an observation through development into a broader design principle.

164. What role does the final paragraph play?

- A) It argues that the fastener did not actually resemble burrs.
- B) It qualifies the idea of copying nature by explaining the need for adaptation.
- C) It repeats only the list of products that use the fastener.
- D) It introduces an unrelated problem with factory inspections.

Literary Passage: The Clockwork Garden

For the library's spring festival, Mei's robotics club built a mechanical garden. Paper flowers were supposed to open one after another as a wooden bee moved along a rail. During the final test, however, the third flower opened before the bee reached it, while the fifth did not open at all.

The team immediately gathered around the fifth flower. Mei stayed near the control box, watching the entire sequence again. The early third flower bothered her more than the silent fifth. If one signal was arriving too soon, she thought, the problem might not belong to either flower.

She slowed the program and displayed each command on the monitor. The labels for flowers three and five had been exchanged. When the program sent the command for the third flower, it activated the fifth flower's motor, which was not yet powered at that point in the sequence. Later, the command labeled for the fifth activated flower three.

After Mei corrected the labels, both flowers worked. The wooden bee traveled down the rail, and the garden unfolded in order.

At the festival, children crowded around the display and tried to guess which flower would open next. Mei listened to their guesses and smiled. The repair had required no new motor and no replaced wire. It had required her to look beyond the part that had failed most visibly.

165. What caused flowers three and five to operate incorrectly?

- A) The control box had no power.
- B) Their command labels had been exchanged.
- C) The wooden bee blocked their wires.
- D) Both motors had burned out.

166. Why does Mei focus on the flower that opens early rather than only on the one that fails to open?

- A) She wants to change the order of the display.
- B) She knows the fifth flower has no motor.
- C) She believes early flowers are more dangerous.
- D) The two unusual behaviors may point to one shared cause.

167. As used in the final paragraph, “visibly” most nearly means

- A) in a way that produces light.
- B) in a way that is easy to notice.
- C) in a way that happens slowly.
- D) in a way that requires a screen.

168. What does the children's attempt to guess the next flower add to the ending?

- A) It suggests that the program remains random and faulty.
- B) It shows the repaired sequence working as an engaging display.
- C) It proves that the wooden bee moves too quickly.
- D) It reveals that the children understand the computer code.

169. Which best describes the passage's structure?

- A) A malfunction leads to observation, diagnosis, repair, and reflection.
- B) Two possible designs are compared without a decision.
- C) A festival is described before a problem is introduced years later.
- D) A machine is built by following a list of numbered instructions.

Informational Passage: Lighting the Night More Carefully

Outdoor lighting can make streets and entrances safer, but poorly directed light also creates glare and brightens the night sky. In many populated areas, this skyglow hides all but the brightest stars. It can also affect animals that rely on darkness for navigation, feeding, or rest.

Reducing unnecessary light does not require leaving useful areas dark. A shielded fixture directs light downward instead of allowing it to spill upward or sideways. A bulb need only be as bright as the task requires. Timers and motion sensors can reduce lighting when no one is present, and warmer-colored light often scatters less in the atmosphere than blue-rich light.

Careful design can improve visibility as well. An extremely bright unshielded lamp may create deep shadows beside the lit area and shine directly into a person's eyes. A lower, evenly directed light can make nearby details easier to see.

Communities that adopt dark-sky practices may begin with public buildings or parking areas, measuring conditions before and after changes. The goal is not darkness for its own sake. It is to use light precisely—where it is needed, when it is needed, and at an appropriate level—while preserving more of the natural night.

170. Which device can reduce lighting when an area is unoccupied?

- A) a blue-rich bulb
- B) a reflective surface
- C) a motion sensor
- D) an unshielded fixture

171. Why can an extremely bright lamp sometimes make nearby details harder to see?

- A) It can create glare and strong contrasts with deep shadows.
- B) It causes motion sensors to stop operating.
- C) It prevents people's eyes from detecting any color.
- D) It always directs all light toward the sky.

172. As used in the final paragraph, “precisely” most nearly means

- A) cheaply and quickly.
- B) carefully and exactly.
- C) brightly and continuously.
- D) naturally and randomly.

173. What is the main idea of the passage?

- A) Brighter bulbs always improve safety and visibility.
- B) Communities should remove every outdoor light to protect stars.
- C) Skyglow is caused mainly by motion sensors and timers.
- D) Outdoor lighting can meet human needs while reducing glare, waste, and harm to the night environment.

174. Which best describes the author's tone?

- A) practical and solution-focused
- B) angry and accusatory
- C) doubtful and confused
- D) humorous and playful

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Which number is greatest?

- A) 0.61
- B) 62%
- C) $\frac{5}{8}$
- D) 0.603

176. What is $3\frac{1}{2} \div \frac{7}{12}$?

- A) 7
- B) 5
- C) 4
- D) 6

177. What is the least common multiple of 16 and 24?

- A) 32
- B) 40
- C) 48
- D) 96

178. What is the greatest common factor of 84 and 126?

- A) 14
- B) 28
- C) 42
- D) 21

179. What is 18% of 250?

- A) 40
- B) 45
- C) 50
- D) 36

180. A value decreases from 90 to 72. What is the percent decrease?

- A) 18%
- B) 22%
- C) 25%
- D) 20%

181. What is the value of $6 + 3(8 - 5)^2$?

- A) 30
- B) 33
- C) 27
- D) 81

182. Which is the prime factorization of 180?

- A) $2^2 \times 3^2 \times 5$
- B) $2 \times 3^2 \times 5$
- C) $2^2 \times 3 \times 5^2$
- D) $2^3 \times 3^2 \times 5$

183. Which fraction is equivalent to 0.375?

- A) $3/8$
- B) $5/12$
- C) $3/10$
- D) $7/16$

184. What is $|-14| + 2$?

- A) 18
- B) 30
- C) 28
- D) 32

185. What is the reciprocal of $2 \frac{2}{3}$?

- A) $2/3$
- B) $5/3$
- C) $3/8$
- D) $8/3$

186. What is the remainder when 157 is divided by 12?

- A) 7
- B) 5
- C) 1
- D) 11

187. Which number is prime?

- A) 81
- B) 87
- C) 77
- D) 83

188. Two quantities are in the ratio 7:9 and have a total of 128. What is the smaller quantity?

- A) 56
- B) 49
- C) 63
- D) 54

189. Which expression is equivalent to $4(2x - 3) + 5$?

- A) $8x - 7$
- B) $6x - 7$
- C) $8x + 2$
- D) $8x - 17$

190. Solve for x: $5x + 11 = 46$.

- A) 9
- B) 7
- C) 5
- D) 6

191. Solve for y: $3(y - 4) = 2y + 9$.

- A) 18
- B) 24
- C) 21
- D) 15

192. Which inequality is equivalent to $6p - 5 \leq 31$?

- A) $p \leq 26/6$
- B) $p \geq 26/6$
- C) $p \geq 6$
- D) $p \leq 6$

193. If $a = 4$ and $b = -3$, what is the value of $2a^2 - b$?

- A) 19
- B) 41
- C) 29
- D) 35

194. The formula $C = 5(F - 32)/9$ converts Fahrenheit temperature F to Celsius temperature C. What is C when F = 68?

- A) 36
- B) 18
- C) 20
- D) 24

195. What is the slope of the line through (2, 5) and (6, 17)?

- A) 3
- B) 6
- C) 4
- D) 2

196. Which equation represents a line with slope 4 and y-intercept -2?

- A) $y = -2x + 4$
- B) $y = 2x - 4$
- C) $y = 4x - 2$
- D) $y = 4x + 2$

197. The first three terms of a sequence are 6, 11, and 16. Which expression gives the nth term?

- A) $5n + 1$
- B) $5n - 1$
- C) $6n - 1$
- D) $6n$

198. A taxi charges a \$4 starting fee plus \$2.50 per mile. Which expression gives the cost for m miles?

- A) $6.50m$
- B) $4m + 2.50$
- C) $2.50(m + 4)$
- D) $2.50m + 4$

199. Twice a number decreased by 7 equals 31. What is the number?

- A) 24
- B) 19
- C) 16
- D) 12

200. What is the solution to the system $x + y = 14$ and $x - y = 4$?

- A) (10, 4)
- B) (9, 5)
- C) (7, 7)
- D) (5, 9)

201. If $f(x) = 3x - 8$, what is $f(7)$?

- A) 15
- B) 29
- C) 13
- D) 19

202. Which value of k makes $2k/3 = 14$?

- A) 24
- B) 21
- C) 28
- D) 18

203. A rectangle's length is 3 cm more than its width. If its perimeter is 38 cm, what is its width?

- A) 7 cm
- B) 8 cm
- C) 9 cm
- D) 11 cm

204. Which expression is equivalent to $(x + 5)(x - 2)$?

- A) $x^2 - 3x - 10$
- B) $x^2 + 3x + 10$
- C) $x^2 + 3x - 10$
- D) $x^2 + 7x + 10$

205. What is the midpoint of the segment joining $(-3, 5)$ and $(7, -1)$?

- A) (2, 2)
- B) (10, -6)
- C) (5, 3)
- D) (4, 4)

206. A right triangle has legs of 8 cm and 15 cm. What is the length of its hypotenuse?

- A) 19 cm
- B) 23 cm
- C) 16 cm
- D) 17 cm

207. What is the area of a trapezoid with bases 10 cm and 16 cm and height 7 cm?

- A) 182 cm^2
- B) 84 cm^2
- C) 91 cm^2
- D) 78 cm^2

208. A regular decagon has a perimeter of 130 cm. What is the length of each side?

- A) 13 cm
- B) 12 cm
- C) 11 cm
- D) 10 cm

209. What is the area of a circle with radius 7 cm?

- A) $28\pi \text{ cm}^2$
- B) $14\pi \text{ cm}^2$
- C) $98\pi \text{ cm}^2$
- D) $49\pi \text{ cm}^2$

210. Each exterior angle of a regular polygon measures 45° . How many sides does the polygon have?

- A) 10
- B) 12
- C) 6
- D) 8

211. An exterior angle of a triangle measures 124° . One remote interior angle measures 57° . What is the other remote interior angle?

- A) 67°
- B) 71°
- C) 63°
- D) 57°

212. What is the sum of the interior angles of a hexagon?

- A) 540°
- B) 720°
- C) 900°
- D) 1080°

213. A rectangle has vertices at (1, 2), (1, 8), (9, 8), and (9, 2). What is its area?

- A) 48 square units
- B) 56 square units
- C) 28 square units
- D) 40 square units

214. A circle has diameter 18 cm. What is its circumference?

- A) 81π cm
- B) 9π cm
- C) 18π cm
- D) 36π cm

215. Which transformation always preserves both length and angle measure?

- A) translation
- B) vertical compression
- C) horizontal stretch
- D) dilation with scale factor 2

216. A square has a diagonal of $10\sqrt{2}$ cm. What is its side length?

- A) 20 cm
- B) 10 cm
- C) $10\sqrt{2}$ cm
- D) 5 cm

217. How many milliliters are in 3.75 liters?

- A) 375
- B) 3,750
- C) 3,075
- D) 750

218. A car travels 156 miles in 3 hours. What is its average speed?

- A) 52 mph
- B) 48 mph
- C) 54 mph
- D) 50 mph

219. A rectangular garden is 14 m long and 9 m wide. How much fencing is needed to enclose it?

- A) 46 m
- B) 23 m
- C) 126 m
- D) 92 m

220. What is the volume of a cylinder with radius 4 cm and height 10 cm?

- A) 40π cm³
- B) 160π cm³
- C) 320π cm³
- D) 80π cm³

221. A rectangular prism measures 6 cm by 8 cm by 11 cm. What is its volume?

- A) 384 cm³
- B) 264 cm³
- C) 528 cm³
- D) 1,056 cm³

222. A 15-foot ladder reaches 12 feet up a wall. How far is the ladder's base from the wall?

- A) 6 ft
- B) 12 ft
- C) 9 ft
- D) 3 ft

223. A room is 12 ft by 15 ft. Carpet costs \$4 per square foot. What is the total carpet cost?

- A) \$480
- B) \$720
- C) \$180
- D) \$108

224. A scale drawing uses 1 cm to represent 2.5 m. A wall measures 7.2 cm on the drawing. What is its actual length?

- A) 20 m
- B) 18 m
- C) 16 m
- D) 22.5 m

225. Water fills $\frac{3}{4}$ of a 28-liter container. How many liters of water are in the container?

- A) 21
- B) 24
- C) 20
- D) 18

226. A clock shows 2:35 p.m. What time will it show 1 hour 50 minutes later?

- A) 4:15 p.m.
- B) 5:25 p.m.
- C) 3:55 p.m.
- D) 4:25 p.m.

227. What is the mean of 12, 15, 18, 21, and 24?

- A) 17
- B) 19
- C) 16
- D) 18

228. What is the median of 7, 11, 13, 18, 22, and 29?

- A) 13
- B) 15.5
- C) 15
- D) 18

229. What is the mode of 4, 6, 6, 7, 8, 8, 8, 10?

- A) 7
- B) 10
- C) 6
- D) 8

230. What is the range of 14, 19, 23, 31, and 38?

- A) 24
- B) 19
- C) 52
- D) 21

231. A bag contains 5 red, 7 blue, and 8 green marbles. What is the probability of selecting a blue marble?

- A) $\frac{2}{5}$
- B) $\frac{7}{13}$
- C) $\frac{7}{20}$
- D) $\frac{1}{4}$

232. A fair number cube is rolled once. What is the probability of rolling a number greater than 4?

- A) $\frac{1}{6}$
- B) $\frac{1}{3}$
- C) $\frac{2}{3}$
- D) $\frac{1}{2}$

233. A spinner has 8 equal sections: 3 yellow, 2 red, and 3 blue. What is the probability of not landing on red?

- A) $\frac{5}{8}$
- B) $\frac{1}{4}$
- C) $\frac{3}{8}$
- D) $\frac{3}{4}$

234. The mean of six numbers is 14. Five of the numbers total 61. What is the sixth number?

- A) 21
- B) 19
- C) 25
- D) 23

235. A survey records 18 votes for art club, 12 for chess club, 15 for drama club, and 5 for music club. What percent chose drama club?

- A) 25%
- B) 15%
- C) 35%
- D) 30%

236. A coin is flipped and a fair number cube is rolled. What is the probability of getting heads and an even number?

- A) $\frac{1}{3}$
- B) $\frac{1}{6}$
- C) $\frac{1}{4}$
- D) $\frac{1}{2}$

237. The lower quartile of a data set is 18 and the upper quartile is 31. What is the interquartile range?

- A) 13
- B) 11
- C) 49
- D) 18

238. Which sample is most likely to represent all students at a middle school?

- A) students leaving basketball practice
- B) members of the student council
- C) students eating in one lunch period
- D) every tenth student on an alphabetized school roster

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which sentence is punctuated correctly?

- A) Before the gates, opened the volunteers checked every table.
- B) Before the gates opened the volunteers checked every table.
- C) Before the gates opened, the volunteers checked every table.
- D) Before, the gates opened the volunteers checked every table.

240. Which sentence uses the apostrophe correctly?

- A) The two hikers packs were stored inside.
- B) The two hikers's packs were stored inside.
- C) The two hikers' packs were stored inside.
- D) The two hiker's packs were stored inside.

241. Which sentence is punctuated correctly?

- A) The kit contains three tools; a wrench, a brush, and a gauge.
- B) The kit contains: three tools, a wrench, a brush, and a gauge.
- C) The kit contains three tools a wrench, a brush, and a gauge.
- D) The kit contains three tools: a wrench, a brush, and a gauge.

242. Which sentence is capitalized correctly?

- A) Our class visited the National Air and Space Museum in April.
- B) Our Class visited the National Air and Space museum in April.
- C) Our class visited the National air and space Museum in April.
- D) Our class visited the national Air and Space Museum in april.

243. Which sentence is capitalized correctly?

- A) Professor lin teaches History at eastview Academy.
- B) Professor Lin teaches History at Eastview academy.
- C) professor Lin teaches history at Eastview Academy.
- D) Professor Lin teaches history at Eastview Academy.

244. Which word is spelled correctly?

- A) rhythym
- B) rhythm
- C) rythm
- D) rythem

245. Which sentence contains a misspelled word?

- A) Her explanation was relevant to the question.
- B) Please separate the paper from the plastic.
- C) The committee reached a unanimous decision.
- D) The guide described the route as managable.

246. Which sentence has correct subject-verb agreement?

- A) The basket of clean towels sits near the door.
- B) The baskets of clean towels sits near the door.
- C) The basket of clean towels are sitting near the door each day.
- D) The basket of clean towels sit near the door.

247. Which sentence uses the pronoun correctly?

- A) The instructor gave Maya and I the maps.
- B) The instructor gave the maps to Maya and me.
- C) The instructor gave she and Maya the maps.
- D) The instructor gave the maps to Maya and myself.

248. Which sentence places the modifier correctly?

- A) The peaches, walking through the orchard, smelled sweet to Lena.
- B) Lena noticed, walking through the orchard, the peaches smelled sweet.
- C) Walking through the orchard, the ripe peaches smelled sweet to Lena.
- D) Walking through the orchard, Lena noticed the sweet smell of ripe peaches.

249. Which sentence is the clearest and most concise?

- A) The final conclusion at the end surprised us.
- B) The conclusion at the final end was surprising to us.
- C) We were surprised by the conclusion that was final.
- D) The conclusion surprised us.

250. Which sentence uses parallel structure?

- A) The role requires to answer calls, keeping accurate records, and visits are scheduled.
- B) The role requires answering calls, accurate records, and to schedule visits.
- C) The role requires calls to be answered, accurate records, and scheduling visits.
- D) The role requires answering calls, keeping accurate records, and scheduling visits.

Editing Passage: Building a School Weather Station

(1) Our science club decided to build a small weather station beside the school garden. (2) Before ordering equipment, we listed the measurements we wanted to collect, including temperature, rainfall, wind speed, and air pressure. (3) The school's oldest classroom has tall windows and a wooden floor. (4) We selected instruments that students could read safely and accurately. (5) Next, we chose an open location away from walls and large trees that might affect the readings. (6) Club members created a schedule so that observations would be recorded at the same time each school day. (7) With consistent measurements, the station will help classes compare daily conditions and identify patterns across the seasons.

251. Which sentence should be removed because it is least relevant to the passage's focus?

- A) Sentence 6
- B) Sentence 3
- C) Sentence 5
- D) Sentence 2

252. Which sentence would be the best topic sentence for the passage?

- A) School gardens can produce vegetables in several seasons.
- B) Building a useful school weather station requires thoughtful equipment, placement, and recordkeeping.
- C) Many scientific instruments contain electronic parts.
- D) Weather changes can make outdoor plans difficult.

253. Where would the following sentence fit best? “This consistency matters because readings taken under different conditions are difficult to compare.”

- A) After sentence 6
- B) After sentence 3
- C) After sentence 5
- D) After sentence 1

254. Which sentence is punctuated correctly?

- A) Nora packed a flashlight; a map; and a first-aid kit.
- B) Nora packed, a flashlight, a map and a first-aid kit.
- C) Nora packed a flashlight, a map, and a first-aid kit.
- D) Nora packed a flashlight a map and, a first-aid kit.

255. Which sentence is punctuated correctly?

- A) The rain stopped; however, the field remained muddy.
- B) The rain stopped: however the field remained muddy.
- C) The rain stopped, however the field remained muddy.
- D) The rain stopped however; the field remained muddy.

256. Which sentence uses quotation marks correctly?

- A) Mina said, The workshop “begins at nine.”
- B) Mina said “The workshop begins at nine”.
- C) Mina said, “The workshop begins at nine.”
- D) Mina said “The workshop begins at nine.”

257. Which sentence is capitalized correctly?

- A) We drove West on Franklin Avenue toward lake Monroe.
- B) We drove West on Franklin avenue toward lake Monroe.
- C) We drove west on Franklin Avenue toward Lake Monroe.
- D) We drove west on franklin Avenue toward Lake monroe.

258. Which sentence is capitalized correctly?

- A) On Monday, aunt Rosa will meet mayor Chen at City hall.
- B) On Monday, Aunt Rosa will meet Mayor Chen at City Hall.
- C) On monday, aunt Rosa will meet Mayor Chen at City Hall.
- D) On Monday, Aunt Rosa will meet Mayor Chen at city hall.

259. Which word is spelled correctly?

- A) embarrass
- B) embarras
- C) embarres
- D) embarass

260. Which sentence contains a misspelled word?

- A) The forecast predicted occasional showers.
- B) The machine requires regular maintenance.
- C) Her response was imediately understood.
- D) The museum displayed an ancient manuscript.

261. Which sentence uses the verb tense correctly?

- A) When Priya reached the station, the train has already depart.
- B) When Priya had reached the station, the train departs tomorrow.
- C) When Priya reaches the station yesterday, the train departed.
- D) When Priya reached the station, the train had already departed.

262. Which sentence correctly uses a comparative adjective?

- A) This path is most narrow than the other one.
- B) This path is narrowest than the other one.
- C) This path is narrower than the other one.
- D) This path is more narrow than the other one.

263. Which sentence is complete?

- A) While the students waited near the entrance.
- B) The guide opened the gate when the group arrived.
- C) Although the map showed another route.
- D) Because the wind became stronger after noon.

264. Which word best completes the sentence? The technician carefully ____ the loose connection before testing the circuit.

- A) performed
- B) located
- C) happened
- D) secured

265. Which sentence avoids redundancy?

- A) We returned back to the starting point.
- B) We went back in return to the starting point.
- C) We returned to the starting point.
- D) We repeated our return back to the starting point.

Editing Passage: Rebuilding a Wetland Boardwalk

(1) A short wooden boardwalk allows visitors to cross the wetland without crushing plants or disturbing muddy soil. (2) After years of rain and winter ice, several boards had weakened. (3) Volunteers first marked the damaged sections and measured each piece that needed replacement. (4) Some wetland birds migrate thousands of miles each year. (5) The team carried materials by hand because heavy vehicles could damage the site. (6) They installed textured boards that would be less slippery when wet. (7) Finally, an ecologist inspected the work and confirmed that construction debris had been removed. (8) The repaired walkway now protects both visitors and the sensitive ground beneath them.

266. Which sentence should be removed because it is least relevant to the boardwalk repair?

- A) Sentence 4
- B) Sentence 5
- C) Sentence 2
- D) Sentence 7

267. Which sentence best states the main idea of the passage?

- A) Volunteer projects should always use heavy construction equipment.
- B) Repairing the boardwalk required careful work that protected the wetland while restoring safe access.
- C) Wetland birds travel long distances during migration.
- D) Wooden walkways are less durable than stone paths.

268. Where would the following sentence fit best? "This preliminary survey allowed the group to bring only the sizes and amounts of lumber required."

- A) After sentence 3
- B) After sentence 1
- C) After sentence 5
- D) After sentence 8

269. Which sentence is punctuated correctly?

- A) If the temperature drops below freezing, cover the young plants.
- B) If the temperature drops, below freezing cover the young plants.
- C) If the temperature drops below freezing; cover the young plants.
- D) If the temperature drops below freezing cover, the young plants.

270. Which sentence uses the apostrophe correctly?

- A) The childrens' projects filled the hallway.
- B) The children's projects filled the hallway.
- C) The childrens projects filled the hallway.
- D) The childrens's projects filled the hallway.

271. Which sentence is capitalized correctly?

- A) My cousin Elena studies French and world history.
- B) My cousin elena studies French and world History.
- C) My cousin Elena studies french and World history.
- D) My Cousin Elena studies french and World History.

272. Which sentence is capitalized correctly?

- A) We read "a Wrinkle in Time" during English class.
- B) We read "A Wrinkle in Time" during English class.
- C) We read "A Wrinkle In Time" during english class.
- D) We read "A wrinkle In time" during english Class.

273. Which word is spelled correctly?

- A) questionnare
- B) questionnaire
- C) questionnaire
- D) quesstionnaire

274. Which word is spelled correctly?

- A) independet
- B) indepanent
- C) independant
- D) independent

275. Which sentence contains a misspelled word?

- A) The experiment produced an unexpected occurence.
- B) The package was delivered to the correct address.
- C) The library acquired several reference books.
- D) The speaker offered a persuasive argument.

276. Which sentence has correct subject-verb agreement?

- A) Neither of the answer are complete.
- B) Neither answers seems complete.
- C) Neither of the answers seem complete.
- D) Neither of the answers seems complete.

277. Which sentence uses the pronoun correctly?

- A) Jordan and him completed the model.
- B) Jordan and himself completed the model.
- C) Him and Jordan completed the model.
- D) Jordan and he completed the model.

278. Which sentence uses the adverb correctly?

- A) The choir sang more beautiful during the ceremony.
- B) The choir beautifully was during the ceremony singing.
- C) The choir sang beautifully during the ceremony.
- D) The choir sang beautiful during the ceremony.

279. Which sentence uses the most precise verb?

- A) The hawk was above the field going.
- B) The hawk did movement above the field.
- C) The hawk went above the field.
- D) The hawk soared above the field.

280. Which sentence uses parallel structure?

- A) The students measured the boards, marked the cuts, and assembled the frame.
- B) The students were measuring boards, marked cuts, and assemble the frame.
- C) The students measured boards, the cuts were marked, and assembling the frame.
- D) The students measured the boards, marking the cuts, and the frame was assembled.

281. Which sentence is clearest?

- A) The coach told Maya, "You will lead the warm-up."
- B) Maya was told by the coach about her leading it.
- C) The coach told Maya that she would lead the warm-up, but it was unclear who she meant.
- D) She told her that she would lead it.

Editing Passage: A Monthly Repair Table

(1) Instead of throwing away small household items, residents can bring them to a monthly repair table at the community center. (2) Volunteers examine loose chair legs, torn fabric, lamps with damaged switches, and other manageable problems. (3) They explain each repair while they work, allowing owners to learn basic skills. (4) Not every object can be fixed safely, so volunteers also help people decide when replacement is necessary. (5) The program reduces waste, shares practical knowledge, and gives neighbors a reason to work together.

282. Which sentence would be the best topic sentence to add before sentence 1?

- A) Volunteers often arrive before a public event begins.
- B) A repair table extends the life of useful objects while teaching practical skills.
- C) Some household items are available in several colors.
- D) Many community centers have large meeting rooms.

283. Which sentence best supports the idea that the program teaches rather than merely fixes?

- A) Sentence 1
- B) Sentence 3
- C) Sentence 5
- D) Sentence 2

284. Which sentence is punctuated correctly?

- A) After, the final bell rang the hallway became quiet.
- B) After the final bell rang, the hallway became quiet.
- C) After the final bell rang; the hallway became quiet.
- D) After the final bell rang the hallway, became quiet.

285. Which sentence is punctuated correctly?

- A) The path was steep: nevertheless the group continued.
- B) The path was steep nevertheless; the group continued.
- C) The path was steep; nevertheless, the group continued.
- D) The path was steep, nevertheless the group continued.

286. Which sentence is capitalized correctly?

- A) In december, Senator Ortiz spoke at Northview Middle school.
- B) In December, Senator ortiz spoke at northview Middle School.
- C) In December, Senator Ortiz spoke at Northview Middle School.
- D) In december, senator Ortiz spoke at Northview middle school.

287. Which sentence is capitalized correctly?

- A) The Mississippi River flows south toward the Gulf of Mexico.
- B) The Mississippi river flows south toward the gulf of Mexico.
- C) The mississippi River flows South toward the Gulf of mexico.
- D) The mississippi river flows South toward the Gulf Of Mexico.

288. Which word is spelled correctly?

- A) perseverance
- B) persevarance
- C) perserverance
- D) perseverance

289. Which word is spelled correctly?

- A) liaison
- B) liason
- C) liaison
- D) leaison

290. Which sentence contains a misspelled word?

- A) The laboratory purchased new equiptment.
- B) The audience remained quiet throughout the performance.
- C) The principal made a formal announcement.
- D) The schedule was flexible enough to change.

291. Which sentence uses the verb tense correctly?

- A) Last week, the class visits the planetarium.
- B) Last week, the class will visit the planetarium.
- C) Last week, the class visited the planetarium.
- D) Last week, the class has visit the planetarium.

292. Which sentence correctly joins the two ideas?

- A) The trail was flooded; so, the hikers chose another route.
- B) The trail was flooded, the hikers chose another route.
- C) The trail was flooded so, the hikers chose another route.
- D) The trail was flooded, so the hikers chose another route.

293. Which sentence clearly indicates that the mountain was covered with fresh snow?

- A) Covered with fresh snow, the hikers admired the mountain.
- B) Covered with fresh snow, the view impressed the hikers near the mountain.
- C) The hikers admired the mountain covered with fresh snow.
- D) The hikers, covered with fresh snow, admired the mountain.

294. Which word best completes the sentence? The new evidence helped ____ the researcher's earlier conclusion.

- A) contain
- B) transfer
- C) perform
- D) confirm

295. Which sentence avoids an unnecessary word?

- A) The two teams joined together in cooperation together.
- B) The two teams cooperated together on the project.
- C) The two teams cooperated on the project.
- D) Together, the two teams mutually cooperated together.

296. Which sentence uses the correct word?

- A) The weather may affect the timing of the event.
- B) The weather may effecting the timing of the event.
- C) The weather may effect the timing of the event.
- D) The weather may have an affect the timing of the event.

Editing Passage: Cataloging a Seed Collection

(1) The garden club stores seeds collected from vegetables and flowers grown on campus. (2) Each envelope must include the plant name, collection date, and garden location. (3) Students also note unusual traits, such as early ripening or strong color. (4) The envelopes are kept in a cool, dry cabinet, where moisture is less likely to damage the seeds. (5) A clear catalog makes it easier to select seeds for future gardens and to compare plants from one season to the next.

297. Which sentence best explains why labels include the garden location?

- A) Some vegetables can be eaten without cooking.
- B) Seeds from the same species may have been grown under different conditions.
- C) Garden tools are stored in a separate cabinet.
- D) Every envelope should be the same color.

298. Where would the following sentence fit best? “Without these details, two similar-looking packets could easily be confused.”

- A) After sentence 5
- B) After sentence 2
- C) After sentence 4
- D) After sentence 1

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. C	2. C	3. C	4. B	5. A	6. D	7. D	8. C
9. A	10. B	11. B	12. D	13. B	14. B	15. C	16. D
17. D	18. D	19. C	20. A	21. C	22. A	23. B	24. D
25. D	26. D	27. C	28. C	29. A	30. B	31. C	32. A
33. B	34. D	35. B	36. A	37. C	38. B	39. D	40. C
41. A	42. B	43. A	44. C	45. A	46. A	47. D	48. B
49. A	50. C	51. A	52. A	53. D	54. B	55. C	56. A
57. D	58. D	59. B	60. B				

Quantitative Skills

61. B	62. C	63. A	64. D	65. A	66. C	67. D	68. C
69. B	70. A	71. C	72. C	73. B	74. A	75. D	76. D
77. A	78. A	79. D	80. B	81. A	82. B	83. D	84. A
85. C	86. C	87. B	88. A	89. D	90. A	91. C	92. A
93. B	94. C	95. B	96. D	97. D	98. C	99. D	100. C
101. D	102. D	103. C	104. B	105. B	106. A	107. B	108. C
109. A	110. B	111. B	112. D				

Reading Comprehension

113. B	114. C	115. A	116. B	117. C	118. B	119. C	120. B
121. C	122. C	123. D	124. D	125. C	126. A	127. A	128. C
129. B	130. B	131. D	132. D	133. D	134. A	135. C	136. A
137. D	138. D	139. D	140. A	141. A	142. D	143. B	144. B
145. C	146. A	147. B	148. C	149. C	150. B	151. A	152. C
153. B	154. D	155. D	156. A	157. A	158. A	159. C	160. A
161. D	162. C	163. D	164. B	165. B	166. D	167. B	168. B
169. A	170. C	171. A	172. B	173. D	174. A		

Mathematics

175. C	176. D	177. C	178. C	179. B	180. D	181. B	182. A
183. A	184. B	185. C	186. C	187. D	188. A	189. A	190. B
191. C	192. D	193. D	194. C	195. A	196. C	197. A	198. D
199. B	200. B	201. C	202. B	203. B	204. C	205. A	206. D

207. C	208. A	209. D	210. D	211. A	212. B	213. A	214. C
215. A	216. B	217. B	218. A	219. A	220. B	221. C	222. C
223. B	224. B	225. A	226. D	227. D	228. B	229. D	230. A
231. C	232. B	233. D	234. D	235. D	236. C	237. A	238. D

Language

239. C	240. C	241. D	242. A	243. D	244. B	245. D	246. A
247. B	248. D	249. D	250. D	251. B	252. B	253. A	254. C
255. A	256. C	257. C	258. B	259. A	260. C	261. D	262. C
263. B	264. D	265. C	266. A	267. B	268. A	269. A	270. B
271. A	272. B	273. B	274. D	275. A	276. D	277. D	278. C
279. D	280. A	281. A	282. B	283. B	284. B	285. C	286. C
287. A	288. D	289. C	290. A	291. C	292. D	293. C	294. D
295. C	296. A	297. B	298. B				

Detailed Explanations

Verbal Skills

1. HSPT-V5-001 - Answer C: precise

Meticulous means showing very careful and precise attention to detail.

2. HSPT-V5-002 - Answer C: relieve

To alleviate a difficulty or pain is to make it less severe, or relieve it.

3. HSPT-V5-003 - Answer C: honest

Candid describes speech that is honest and direct.

4. HSPT-V5-004 - Answer B: adaptable

A resilient person or thing can recover and adapt after difficulty.

5. HSPT-V5-005 - Answer A: brief

Concise language expresses an idea clearly in few words, so brief is closest.

6. HSPT-V5-006 - Answer D: outdated

Obsolete describes something no longer used because it has been replaced or is outdated.

7. HSPT-V5-007 - Answer D: passionate

Fervent means displaying intense feeling or enthusiasm.

8. HSPT-V5-008 - Answer C: fair

Impartial means fair and not favoring one side.

9. HSPT-V5-009 - Answer A: uncertain

A precarious situation is unstable or uncertain and may become dangerous.

10. HSPT-V5-010 - Answer B: increase

To augment something is to add to it or increase it.

11. HSPT-V5-011 - Answer B: peaceful

Tranquil means calm and peaceful.

12. HSPT-V5-012 - Answer D: recognize

To discern something is to perceive or recognize it clearly.

13. HSPT-V5-013 - Answer B: scarce

Abundant means plentiful, while scarce means available only in a small amount.

14. HSPT-V5-014 - Answer B: flexible

Rigid means stiff or inflexible, so flexible is its opposite.

15. HSPT-V5-015 - Answer C: opaque

Transparent material lets light through; opaque material does not.

16. HSPT-V5-016 - Answer D: eager

Reluctant means unwilling, whereas eager means strongly willing.

17. HSPT-V5-017 - Answer D: expert

A novice is a beginner, while an expert has extensive skill or knowledge.

18. HSPT-V5-018 - Answer D: contract

Expand means become larger; contract means become smaller.

19. HSPT-V5-019 - Answer C: temporary

Permanent means lasting indefinitely, while temporary means lasting a limited time.

20. HSPT-V5-020 - Answer A: rude

Courteous behavior is polite; rude behavior is impolite.

21. HSPT-V5-021 - Answer C: natural

Artificial means made rather than occurring naturally.

22. HSPT-V5-022 - Answer A: pessimistic

Optimistic expects favorable outcomes; pessimistic expects unfavorable ones.

23. HSPT-V5-023 - Answer B: specific

Vague means unclear or imprecise, while specific means exact and clearly identified.

24. HSPT-V5-024 - Answer D: friendly
Hostile means unfriendly or opposed, so friendly is the opposite.
25. HSPT-V5-025 - Answer D: heartbeats
A seismograph detects and records earthquake vibrations, while a stethoscope detects sounds such as heartbeats.
26. HSPT-V5-026 - Answer D: meal
A blueprint guides the construction of a building, just as a recipe guides preparation of a meal.
27. HSPT-V5-027 - Answer C: weight
An odometer measures distance, while a scale measures weight.
28. HSPT-V5-028 - Answer C: eagle
A cub is a young bear, and an eaglet is a young eagle.
29. HSPT-V5-029 - Answer A: conductor
A judge characteristically uses a gavel, while a conductor uses a baton.
30. HSPT-V5-030 - Answer B: fish
A petal forms part of a flower, and a scale forms part of a fish's outer covering.
31. HSPT-V5-031 - Answer C: bright
A whisper is notably quiet, while a glare of light is notably bright.
32. HSPT-V5-032 - Answer A: interpret
An architect designs, while a translator interprets language.
33. HSPT-V5-033 - Answer B: constellation
An archipelago is a recognized group of islands, and a constellation is a recognized group of stars.
34. HSPT-V5-034 - Answer D: animal
Sap circulates within a tree, while blood circulates within an animal.
35. HSPT-V5-035 - Answer B: stingy
Cautious and reckless are opposites, just as generous and stingy are opposites.
36. HSPT-V5-036 - Answer A: symphony
A chapter is a major division of a novel, and a movement is a major division of a symphony.
37. HSPT-V5-037 - Answer C: copper
Basalt, marble, and limestone are rocks; copper is a metal.
38. HSPT-V5-038 - Answer B: oak
Elm, birch, maple, and oak are deciduous trees.
39. HSPT-V5-039 - Answer D: sphere
Triangle, rectangle, and pentagon are plane polygons; a sphere is three-dimensional.
40. HSPT-V5-040 - Answer C: harp
All four are string instruments; the other choices are wind instruments.
41. HSPT-V5-041 - Answer A: applaud
Whisper, murmur, and mumble are ways of speaking quietly; applaud is clapping.
42. HSPT-V5-042 - Answer B: July
January, March, May, and July each have 31 days.
43. HSPT-V5-043 - Answer A: turquoise
Scarlet, crimson, and maroon are shades of red; turquoise is blue-green.
44. HSPT-V5-044 - Answer C: journal
Biography, memoir, autobiography, and journal are nonfiction forms centered on lived experience.
45. HSPT-V5-045 - Answer A: erode
Evaporate, condense, and freeze are changes of state; erode is the wearing away of material.
46. HSPT-V5-046 - Answer A: plumber
All four are skilled building trades.
47. HSPT-V5-047 - Answer D: sirius
Mercury, Venus, and Saturn are planets; Sirius is a star.
48. HSPT-V5-048 - Answer B: second
Meter, liter, gram, and second are units of measurement.
49. HSPT-V5-049 - Answer A: No cedar box is transparent.
Because every cedar box is wooden and no wooden object is transparent, no cedar box can be transparent.

50. HSPT-V5-050 - Answer C: Jade

The required order is Jade, Mina, Owen, Luis, so Jade must finish first.

51. HSPT-V5-051 - Answer A: Caleb attends Wednesday.

Since the only choices are Wednesday and Saturday and Caleb cannot attend Saturday, he must attend Wednesday.

52. HSPT-V5-052 - Answer A: The library is not open.

The contrapositive of the rule is that if the lights are not on, the library is not open.

53. HSPT-V5-053 - Answer D: Nora

The comparisons establish Nora above Eli, Eli above Priya, and Priya above Sam.

54. HSPT-V5-054 - Answer B: Talia attends practice.

The general rule applies directly to Talia because she is a team member.

55. HSPT-V5-055 - Answer C: R47

R47 has one letter followed by two different digits and begins with R4.

56. HSPT-V5-056 - Answer A: robotics

Cole has chess; Ben cannot have robotics and therefore has art, leaving robotics for Ana.

57. HSPT-V5-057 - Answer D: red, green, blue

This order places green immediately after red and blue somewhere to red's right.

58. HSPT-V5-058 - Answer D: No lizards have feathers.

Every lizard is a reptile, and reptiles have no feathers, so lizards have no feathers.

59. HSPT-V5-059 - Answer B: Oak

The only possible order is Pine, Oak, River, Lake.

60. HSPT-V5-060 - Answer B: Shape X is not a square.

By the contrapositive, a shape without four equal sides cannot be a square.

Quantitative Skills

61. HSPT-Q5-001 - Answer B: 159

Each term is twice the preceding term plus 1, so $79 \times 2 + 1 = 159$.

62. HSPT-Q5-002 - Answer C: 36

The sequence subtracts 14, 13, 12, 11, and then 10; $46 - 10 = 36$.

63. HSPT-Q5-003 - Answer A: 34

The operations alternate multiplying by 2 and subtracting 1; $17 \times 2 = 34$.

64. HSPT-Q5-004 - Answer D: 50

The increases are 3, 6, 9, and 12; the next increase is 15, giving 50.

65. HSPT-Q5-005 - Answer A: 56

The terms are products of consecutive integers: 2×3 , 3×4 , ..., so the next is $7 \times 8 = 56$.

66. HSPT-Q5-006 - Answer C: 3

Each term is one third of the preceding term, so $9 \div 3 = 3$.

67. HSPT-Q5-007 - Answer D: 52

The increases are 3, 6, 9, and 12; adding 15 next gives 52.

68. HSPT-Q5-008 - Answer C: 94

Each term is twice the previous term plus 2, so $46 \times 2 + 2 = 94$.

69. HSPT-Q5-009 - Answer B: 4

Each term is half the preceding term, so half of 8 is 4.

70. HSPT-Q5-010 - Answer A: 16.2

Each term is three times the preceding term, so $5.4 \times 3 = 16.2$.

71. HSPT-Q5-011 - Answer C: 135

The increases double: 4, 8, 16, 32, then 64; $71 + 64 = 135$.

72. HSPT-Q5-012 - Answer C: 23

The increases are consecutive integers 1 through 5, so adding 6 gives 23.

73. HSPT-Q5-013 - Answer B: 55

The decreases are 3, 6, 9, and 12; subtracting 15 next gives 55.

74. HSPT-Q5-014 - Answer A: 97

Each term is twice the preceding term minus 1; $49 \times 2 - 1 = 97$.

75. HSPT-Q5-015 - Answer D: 9

If $4x + 9 = 45$, then $4x = 36$ and $x = 9$.

76. HSPT-Q5-016 - Answer D: 24

37.5% equals $\frac{3}{8}$, and $\frac{3}{8}$ of 64 is 24.

77. HSPT-Q5-017 - Answer A: 24

A scale factor of 8 changes 5 blue parts to 40, so the 3 white parts equal $3 \times 8 = 24$.

78. HSPT-Q5-018 - Answer A: 2

Convert to improper fractions: $\frac{13}{5} \div \frac{13}{10} = \frac{13}{5} \times \frac{10}{13} = 2$.

79. HSPT-Q5-019 - Answer D: 28

Four numbers with mean 24 total 96; the known numbers total 68, so $n = 28$.

80. HSPT-Q5-020 - Answer B: \$54

A 25% discount is \$18, so the sale price is $\$72 - \$18 = \$54$.

81. HSPT-Q5-021 - Answer A: \$10.50

Each notebook costs $\$24.50 \div 7 = \3.50 , and 3 cost \$10.50.

82. HSPT-Q5-022 - Answer B: $3(6) - 1$

Evaluating gives $18 - 1 = 17$; the other expressions equal 18, 19, and 19.

83. HSPT-Q5-023 - Answer D: 60 liters

The added 18 liters represent $\frac{9}{10} - \frac{3}{5} = \frac{3}{10}$ of the tank, so capacity is $18 \div \frac{3}{10} = 60$ liters.

84. HSPT-Q5-024 - Answer A: 180

The least common multiple uses 2^2 , 3^2 , and 5, giving $4 \times 9 \times 5 = 180$.

85. HSPT-Q5-025 - Answer C: 63 miles
Multiply 3.5 inches by 18 miles per inch to obtain 63 miles.
86. HSPT-Q5-026 - Answer C: 21
Solve $2x + 8 = 50$: $2x = 42$ and $x = 21$.
87. HSPT-Q5-027 - Answer B: $3 \frac{3}{4}$
Multiply $2 \frac{1}{4}$ by $10/6$: $9/4 \times 5/3 = 15/4 = 3 \frac{3}{4}$ cups.
88. HSPT-Q5-028 - Answer A: 9
The absolute value is 18 and $\sqrt{81}$ is 9, so $18 - 9 = 9$.
89. HSPT-Q5-029 - Answer D: The two quantities are equal.
Quantity A is 35 because $60 \div 12 \times 7 = 35$, matching Quantity B.
90. HSPT-Q5-030 - Answer A: Quantity A is greater.
Quantity A is 36, while Quantity B is 33, so Quantity A is greater.
91. HSPT-Q5-031 - Answer C: The two quantities are equal.
Solving $5x - 7 = 48$ gives $5x = 55$ and $x = 11$.
92. HSPT-Q5-032 - Answer A: Quantity A is greater.
The greatest common factor is 14, which is greater than 12.
93. HSPT-Q5-033 - Answer B: The two quantities are equal.
Both quantities equal 150.
94. HSPT-Q5-034 - Answer C: The two quantities are equal.
Quantity A is 36, and Quantity B is $9 \times 4 = 36$.
95. HSPT-Q5-035 - Answer B: The two quantities are equal.
Both divisions leave a remainder of 1.
96. HSPT-Q5-036 - Answer D: The two quantities are equal.
The median is 12, and the mean is $(8 + 12 + 16) \div 3 = 12$.
97. HSPT-Q5-037 - Answer D: The two quantities are equal.
Both quantities equal 23.
98. HSPT-Q5-038 - Answer C: The two quantities are equal.
The fraction $5/8$ converts to 0.625.
99. HSPT-Q5-039 - Answer D: Quantity A is greater.
Solving gives $y/3 = 6$ and $y = 18$, which is greater than 15.
100. HSPT-Q5-040 - Answer C: Quantity A is greater.
The positive factors of 24 are 1, 2, 3, 4, 6, 8, 12, and 24, for a total of 8.
101. HSPT-Q5-041 - Answer D: Quantity A is greater.
The semicircle's area is $1/2 \times 3.14 \times 36 = 56.52 \text{ cm}^2$, while the triangle's area is 56 cm^2 .
102. HSPT-Q5-042 - Answer D: The two quantities are equal.
Both perimeters equal 36 cm.
103. HSPT-Q5-043 - Answer C: Quantity A is greater.
The cube's volume is $5^3 = 125 \text{ cm}^3$, which is greater than 120 cm^3 .
104. HSPT-Q5-044 - Answer B: Quantity A is greater.
The diameter is twice the radius, or 26 cm.
105. HSPT-Q5-045 - Answer B: The two quantities are equal.
The third angle is $180^\circ - 52^\circ - 71^\circ = 57^\circ$.
106. HSPT-Q5-046 - Answer A: The two quantities are equal.
The triangle's area is $1/2 \times 15 \times 8 = 60 \text{ cm}^2$, and the parallelogram's area is $10 \times 6 = 60 \text{ cm}^2$.
107. HSPT-Q5-047 - Answer B: Quantity A is greater.
The circumference is $2 \times 3.14 \times 4 = 25.12 \text{ cm}$, slightly greater than 25 cm.
108. HSPT-Q5-048 - Answer C: The two quantities are equal.
The hexagon's perimeter is $6 \times 6 = 36 \text{ cm}$ and the square's is $4 \times 9 = 36 \text{ cm}$.
109. HSPT-Q5-049 - Answer A: Quantity A is greater.
The prism's volume is $3 \times 5 \times 7 = 105 \text{ cm}^3$.
110. HSPT-Q5-050 - Answer B: Quantity B is greater.
The circle's area is $3.14 \times 3^2 = 28.26 \text{ cm}^2$, less than 30 cm^2 .

111. HSPT-Q5-051 - Answer B: Quantity B is greater.

A rectangle has 2 diagonals, while a pentagon has 5.

112. HSPT-Q5-052 - Answer D: The two quantities are equal.

The interior angles of every quadrilateral sum to 360° .

Reading Comprehension

113. HSPT-R5-001 - Answer B: The antenna appeared near the edge of the crosshairs.
The passage states that the antenna was near the edge rather than centered in the crosshairs.
114. HSPT-R5-002 - Answer C: He wants Leila to learn the alignment process.
He guides Leila with a question and tools so that she can understand and perform the correction herself.
115. HSPT-R5-003 - Answer A: shake slightly.
Atmospheric movement makes the image appear to shake or waver slightly.
116. HSPT-R5-004 - Answer B: Understanding a process can be as rewarding as obtaining the result.
Leila values not only the clear view but also knowing why her adjustments produced it.
117. HSPT-R5-005 - Answer C: She has become more patient and deliberate.
Her controlled adjustments contrast with her hurried first attempt and show growth in patience.
118. HSPT-R5-006 - Answer B: structures produced by cultivated fungus
The first and third paragraphs explain that the ants cultivate fungus as their main food.
119. HSPT-R5-007 - Answer C: They help regulate conditions by circulating air.
The passage connects airflow through tunnels with the temperature and moisture needed in the gardens.
120. HSPT-R5-008 - Answer B: move into different locations.
The ants carry material underground and excavate soil, moving nutrients from one place to another.
121. HSPT-R5-009 - Answer C: To explain how leafcutter ants cultivate fungus and affect their environment
The passage describes the farming partnership and the colony's broader effects on soil.
122. HSPT-R5-010 - Answer C: It broadens the focus from the colony's farm to its environmental effects.
The passage shifts from how the farm works to how colony activity changes vegetation and soil.
123. HSPT-R5-011 - Answer D: repeating units joined into a larger design
The first paragraph defines blocks as repeating units that are joined to form a larger design.
124. HSPT-R5-012 - Answer D: Part of the fabric will be taken up by seam allowances.
The passage explains that seam allowances disappear inside seams, reducing the visible size.
125. HSPT-R5-013 - Answer C: To evaluate the overall balance from a distance
The passage says this method allows the quilter to step back, notice imbalance, and rearrange pieces.
126. HSPT-R5-014 - Answer A: intentional.
A deliberate break in symmetry is one created intentionally to attract attention.
127. HSPT-R5-015 - Answer A: Mathematical structure contributes to the practical construction and artistry of quilts.
The passage connects geometry and measurement with both construction and creative expression.
128. HSPT-R5-016 - Answer C: It emphasizes that precision supports rather than limits creativity.
The phrase presents accurate structure as a framework that enables artistic choices.
129. HSPT-R5-017 - Answer B: The two diagonal measurements are different.
Malik discovers that one diagonal is half an inch longer than the other.
130. HSPT-R5-018 - Answer B: She wants him to investigate the cause himself.
By providing measuring tools and stepping away, she encourages Malik to diagnose the frame.
131. HSPT-R5-019 - Answer D: moved slightly.
Malik makes small adjustments to the boards until the diagonal measurements match.
132. HSPT-R5-020 - Answer D: A construction problem is discovered, investigated, corrected, and reconsidered.
The narrative follows Malik from the rocking frame through diagnosis, repair, and reflection.
133. HSPT-R5-021 - Answer D: The stable bench demonstrates the value of the accurate measurement.
No one sees the measurement itself, but the bench's stability shows its practical effect.
134. HSPT-R5-022 - Answer A: an anemone
The second paragraph specifically states that anemones close their bodies to reduce water loss.
135. HSPT-R5-023 - Answer C: They remain cut off from fresh seawater for longer periods.
Longer separation gives evaporation or rainfall more time to change the concentration of salt.
136. HSPT-R5-024 - Answer A: Tide pools are variable habitats whose organisms use different strategies to survive.
The passage centers on changing conditions, adaptations, and differences by location.

137. HSPT-R5-025 - Answer D: To emphasize that conditions vary through interacting influences
The description summarizes how sunlight, rain, evaporation, tides, and waves continually affect the habitat.
138. HSPT-R5-026 - Answer D: It challenges a first impression and introduces hidden difficulty.
The contrast prepares readers to look beyond the calm appearance to rapidly changing conditions.
139. HSPT-R5-027 - Answer D: what people ate and when they may have occupied a site
The second paragraph identifies diet and seasonal occupation as possible evidence from bones.
140. HSPT-R5-028 - Answer A: Layers can be disturbed or mixed over time.
The passage warns that digging, floods, or animals can mix deposits from different times.
141. HSPT-R5-029 - Answer A: Reports may reflect general intentions while trash records exceptions too.
The passage notes that people may describe usual behavior while the physical record includes deviations.
142. HSPT-R5-030 - Answer D: decompose.
Wood and cloth may break down naturally over time, which is to decompose.
143. HSPT-R5-031 - Answer B: To explain how discarded materials can reveal daily behavior when interpreted carefully
The passage describes evidence from refuse and repeatedly explains the limits on interpreting it.
144. HSPT-R5-032 - Answer B: To show that discarded materials can reveal behavior in different periods
The two examples extend the same principle across past and present communities.
145. HSPT-R5-033 - Answer C: She restarts the audio program.
The passage states that Ms. Reed restarts the program while Jonah fills the silence.
146. HSPT-R5-034 - Answer A: He wants space for a backup idea if the planned sequence fails.
The sentence on the card records the lesson that he should keep the program moving when a plan fails.
147. HSPT-R5-035 - Answer B: gradually lowered the sound from.
In radio production, fading down a microphone means reducing its audio level gradually.
148. HSPT-R5-036 - Answer C: Preparation is valuable, but adaptability is also necessary when problems occur.
Jonah is prepared, yet success depends on improvising when the recording fails.
149. HSPT-R5-037 - Answer C: It emphasizes the sudden failure and moment of suspense.
The short isolated sentence interrupts the smooth action just as the failed recording interrupts the program.
150. HSPT-R5-038 - Answer B: sugars made using sunlight
The first paragraph explains that the alga or bacterium uses sunlight to produce sugars.
151. HSPT-R5-039 - Answer A: They absorb atmospheric substances and some respond to pollution.
Their direct exposure to the atmosphere and differing sensitivity make their presence biologically informative.
152. HSPT-R5-040 - Answer C: begin to grow successfully.
Mosses and plants take hold when they become established and begin growing.
153. HSPT-R5-041 - Answer B: Lichens are partnerships that survive in difficult places and contribute to environmental change.
The passage explains lichen structure, survival, rock breakdown, and value as environmental indicators.
154. HSPT-R5-042 - Answer D: It moves from defining lichens to explaining their roles in landscapes and environmental study.
The scope broadens from the partnership itself to weathering, soil formation, and pollution monitoring.
155. HSPT-R5-043 - Answer D: They invite narrators to give fuller accounts.
The passage contrasts an open prompt with a yes-or-no question and says the former produces richer responses.
156. HSPT-R5-044 - Answer A: To check factual details while still studying the narrator's perspective
The passage distinguishes perspective from facts that may need confirmation elsewhere.
157. HSPT-R5-045 - Answer A: private or potentially harmful if disclosed.
Sensitive portions require restrictions because sharing them could invade privacy or cause harm.
158. HSPT-R5-046 - Answer A: Oral history preserves personal accounts and voices but requires careful questioning and interpretation.
The passage explains interviewing, memory limits, archival choices, and the value of vocal expression.
159. HSPT-R5-047 - Answer C: To illustrate qualities of spoken testimony that a summary may not capture
The list supplies concrete examples of meaning carried by a recorded voice.
160. HSPT-R5-048 - Answer A: Tiny hooks catch in loops of fiber or hair.
The first paragraph identifies the hook-and-loop interaction observed under magnification.

161. HSPT-R5-049 - Answer D: A natural principle had to be adapted into a durable, manufacturable product. The passage describes challenges in flexibility, strength, repeated use, and uniform manufacturing.
162. HSPT-R5-050 - Answer C: Close observation of nature can inspire technology, but practical design requires further problem solving.
De Mestral's observation began the idea, but years of adaptation made it usable.
163. HSPT-R5-051 - Answer D: It traces an observation through development into a broader design principle. The text moves from burrs to the fastener's development and then to biomimicry.
164. HSPT-R5-052 - Answer B: It qualifies the idea of copying nature by explaining the need for adaptation. The final paragraph explains that biomimicry uses a principle but must meet different human requirements.
165. HSPT-R5-053 - Answer B: Their command labels had been exchanged.
Mei discovers that the labels for the two flowers are reversed in the program.
166. HSPT-R5-054 - Answer D: The two unusual behaviors may point to one shared cause.
Her suspicion leads her to inspect the sequence and discover that the two commands are swapped.
167. HSPT-R5-055 - Answer B: in a way that is easy to notice.
The fifth flower's failure is the most obvious or noticeable symptom.
168. HSPT-R5-056 - Answer B: It shows the repaired sequence working as an engaging display.
Their anticipation confirms both the orderly function and the audience appeal of the repaired garden.
169. HSPT-R5-057 - Answer A: A malfunction leads to observation, diagnosis, repair, and reflection.
The narrative follows a clear problem-solving sequence from failed test to successful festival.
170. HSPT-R5-058 - Answer C: a motion sensor
The second paragraph lists timers and motion sensors as ways to reduce light when no one is present.
171. HSPT-R5-059 - Answer A: It can create glare and strong contrasts with deep shadows.
The passage explains that glare and dark adjacent shadows can reduce useful visibility.
172. HSPT-R5-060 - Answer B: carefully and exactly.
The goal is to direct the appropriate amount of light exactly where and when it is needed.
173. HSPT-R5-061 - Answer D: Outdoor lighting can meet human needs while reducing glare, waste, and harm to the night environment.
The passage presents practical ways to use light effectively while limiting unwanted effects.
174. HSPT-R5-062 - Answer A: practical and solution-focused
The author acknowledges the need for lighting and explains specific improvements rather than rejecting it entirely.

Mathematics

175. HSPT-M5-001 - Answer C: $\frac{5}{8}$

The values are 0.625, 0.61, 0.62, and 0.603, so $\frac{5}{8}$ is greatest.

176. HSPT-M5-002 - Answer D: 6

Convert $3\frac{1}{2}$ to $\frac{7}{2}$, then multiply by $\frac{12}{7}$: $\frac{7}{2} \times \frac{12}{7} = 6$.

177. HSPT-M5-003 - Answer C: 48

The smallest positive number divisible by both 16 and 24 is 48.

178. HSPT-M5-004 - Answer C: 42

Both numbers are divisible by 42, and no larger positive factor divides both.

179. HSPT-M5-005 - Answer B: 45

Multiply 250 by 0.18 to obtain 45.

180. HSPT-M5-006 - Answer D: 20%

The decrease is 18, and $18 \div 90 = 0.20$, or 20%.

181. HSPT-M5-007 - Answer B: 33

Evaluate parentheses and exponent first: $6 + 3(3^2) = 6 + 27 = 33$.

182. HSPT-M5-008 - Answer A: $2^2 \times 3^2 \times 5$

Dividing 180 into prime factors gives $2 \times 2 \times 3 \times 3 \times 5$.

183. HSPT-M5-009 - Answer A: $\frac{3}{8}$

$0.375 = \frac{375}{1000}$, which simplifies to $\frac{3}{8}$.

184. HSPT-M5-010 - Answer B: 30

The absolute value is 14 and $2\square$ is 16; their sum is 30.

185. HSPT-M5-011 - Answer C: $\frac{3}{8}$

The mixed number is $\frac{8}{3}$, whose reciprocal is $\frac{3}{8}$.

186. HSPT-M5-012 - Answer C: 1

Twelve times 13 is 156, leaving a remainder of 1.

187. HSPT-M5-013 - Answer D: 83

The number 83 has no positive factors other than 1 and 83.

188. HSPT-M5-014 - Answer A: 56

There are 16 ratio parts; each is $128 \div 16 = 8$, and $7 \times 8 = 56$.

189. HSPT-M5-015 - Answer A: $8x - 7$

Distribute 4 to get $8x - 12$, then add 5 to obtain $8x - 7$.

190. HSPT-M5-016 - Answer B: 7

Subtract 11 to get $5x = 35$, then divide by 5.

191. HSPT-M5-017 - Answer C: 21

Expand to $3y - 12 = 2y + 9$; subtract $2y$ and add 12 to get $y = 21$.

192. HSPT-M5-018 - Answer D: $p \leq 6$

Add 5 to obtain $6p \leq 36$, then divide by 6.

193. HSPT-M5-019 - Answer D: 35

Substitute: $2(4^2) - (-3) = 32 + 3 = 35$.

194. HSPT-M5-020 - Answer C: 20

Substitute 68: $5(68 - 32)/9 = 5 \times 36/9 = 20$.

195. HSPT-M5-021 - Answer A: 3

Slope is the change in y divided by the change in x : $(17 - 5)/(6 - 2) = 12/4 = 3$.

196. HSPT-M5-022 - Answer C: $y = 4x - 2$

Slope-intercept form is $y = mx + b$, so $m = 4$ and $b = -2$.

197. HSPT-M5-023 - Answer A: $5n + 1$

The common difference is 5, and $5(1) + 1 = 6$, so the rule is $5n + 1$.

198. HSPT-M5-024 - Answer D: $2.50m + 4$

The variable cost is \$2.50 times the miles, and the fixed starting fee adds \$4.

199. HSPT-M5-025 - Answer B: 19
The equation $2x - 7 = 31$ gives $2x = 38$ and $x = 19$.
200. HSPT-M5-026 - Answer B: (9, 5)
Adding the equations gives $2x = 18$, so $x = 9$ and $y = 5$.
201. HSPT-M5-027 - Answer C: 13
Substitute 7: $f(7) = 3(7) - 8 = 21 - 8 = 13$.
202. HSPT-M5-028 - Answer B: 21
Multiply both sides by $3/2$: $k = 14 \times 3/2 = 21$.
203. HSPT-M5-029 - Answer B: 8 cm
Let width be w and length $w + 3$. Then $2w + 2(w + 3) = 38$, so $4w = 32$ and $w = 8$.
204. HSPT-M5-030 - Answer C: $x^2 + 3x - 10$
Multiply the binomials: $x^2 - 2x + 5x - 10 = x^2 + 3x - 10$.
205. HSPT-M5-031 - Answer A: (2, 2)
Average the x-coordinates and y-coordinates: $((-3 + 7)/2, (5 - 1)/2) = (2, 2)$.
206. HSPT-M5-032 - Answer D: 17 cm
By the Pythagorean theorem, $\sqrt{(8^2 + 15^2)} = \sqrt{289} = 17$.
207. HSPT-M5-033 - Answer C: 91 cm²
Area is $1/2(b_1 + b_2)h = 1/2(26)(7) = 91$ cm².
208. HSPT-M5-034 - Answer A: 13 cm
A decagon has 10 equal sides, so each side is $130 \div 10 = 13$ cm.
209. HSPT-M5-035 - Answer D: 49π cm²
Circle area is πr^2 , so $\pi(7^2) = 49\pi$ cm².
210. HSPT-M5-036 - Answer D: 8
The exterior angles of a polygon total 360° , so the number of sides is $360 \div 45 = 8$.
211. HSPT-M5-037 - Answer A: 67°
An exterior angle equals the sum of the remote interior angles, so $124^\circ - 57^\circ = 67^\circ$.
212. HSPT-M5-038 - Answer B: 720°
For a hexagon, $(6 - 2) \times 180^\circ = 720^\circ$.
213. HSPT-M5-039 - Answer A: 48 square units
Its dimensions are 8 units by 6 units, so the area is 48 square units.
214. HSPT-M5-040 - Answer C: 18π cm
Circumference equals π times the diameter, so it is 18π cm.
215. HSPT-M5-041 - Answer A: translation
A translation is a rigid motion, so it preserves all lengths and angle measures.
216. HSPT-M5-042 - Answer B: 10 cm
A square's diagonal equals side $\times \sqrt{2}$, so the side is 10 cm.
217. HSPT-M5-043 - Answer B: 3,750
One liter is 1,000 milliliters, so 3.75 liters equals 3,750 milliliters.
218. HSPT-M5-044 - Answer A: 52 mph
Average speed is distance divided by time: $156 \div 3 = 52$ mph.
219. HSPT-M5-045 - Answer A: 46 m
The perimeter is $2(14 + 9) = 46$ m.
220. HSPT-M5-046 - Answer B: 160π cm³
Cylinder volume is $\pi r^2 h = \pi(4^2)(10) = 160\pi$ cm³.
221. HSPT-M5-047 - Answer C: 528 cm³
Multiply the three dimensions: $6 \times 8 \times 11 = 528$ cm³.
222. HSPT-M5-048 - Answer C: 9 ft
The wall, ground, and ladder form a right triangle: $\sqrt{(15^2 - 12^2)} = \sqrt{81} = 9$ ft.
223. HSPT-M5-049 - Answer B: \$720
The area is $12 \times 15 = 180$ square feet, and $180 \times \$4 = \720 .
224. HSPT-M5-050 - Answer B: 18 m
Multiply 7.2 by 2.5 to get 18 meters.

225. HSPT-M5-051 - Answer A: 21
Three fourths of 28 is $28 \div 4 \times 3 = 21$ liters.
226. HSPT-M5-052 - Answer D: 4:25 p.m.
Adding 1 hour gives 3:35 p.m.; adding 50 minutes gives 4:25 p.m.
227. HSPT-M5-053 - Answer D: 18
The sum is 90, and $90 \div 5 = 18$.
228. HSPT-M5-054 - Answer B: 15.5
The middle values are 13 and 18, whose average is 15.5.
229. HSPT-M5-055 - Answer D: 8
The value 8 occurs three times, more than any other value.
230. HSPT-M5-056 - Answer A: 24
Range is maximum minus minimum: $38 - 14 = 24$.
231. HSPT-M5-057 - Answer C: 7/20
There are 20 marbles total and 7 are blue, so the probability is 7/20.
232. HSPT-M5-058 - Answer B: 1/3
The favorable results are 5 and 6, so the probability is $2/6 = 1/3$.
233. HSPT-M5-059 - Answer D: 3/4
Six of eight sections are not red, so the probability is $6/8 = 3/4$.
234. HSPT-M5-060 - Answer D: 23
All six numbers total $6 \times 14 = 84$; $84 - 61 = 23$.
235. HSPT-M5-061 - Answer D: 30%
There are 50 votes total, and $15/50 = 30\%$.
236. HSPT-M5-062 - Answer C: 1/4
The events are independent: $1/2 \times 3/6 = 1/4$.
237. HSPT-M5-063 - Answer A: 13
Interquartile range is $Q3 - Q1 = 31 - 18 = 13$.
238. HSPT-M5-064 - Answer D: every tenth student on an alphabetized school roster
A systematic sample from the full roster is less biased than samples drawn from particular activities or schedules.

Language

239. HSPT-L5-001 - Answer C: Before the gates opened, the volunteers checked every table.
A comma follows the introductory dependent clause before the independent clause.
240. HSPT-L5-002 - Answer C: The two hikers' packs were stored inside.
The packs belong to two hikers, so the plural possessive hikers' is correct.
241. HSPT-L5-003 - Answer D: The kit contains three tools: a wrench, a brush, and a gauge.
A colon correctly follows a complete clause that introduces a list.
242. HSPT-L5-004 - Answer A: Our class visited the National Air and Space Museum in April.
The official museum name and the month are proper nouns and require capitals; class does not.
243. HSPT-L5-005 - Answer D: Professor Lin teaches history at Eastview Academy.
A title before a name and a complete school name are capitalized, while a general subject is lowercase.
244. HSPT-L5-006 - Answer B: rhythm
Rhythm is spelled r-h-y-t-h-m.
245. HSPT-L5-007 - Answer D: The guide described the route as manageable.
Manageable is the correct spelling; managable is missing the e after g.
246. HSPT-L5-008 - Answer A: The basket of clean towels sits near the door.
The singular subject basket takes the singular verb sits.
247. HSPT-L5-009 - Answer B: The instructor gave the maps to Maya and me.
Me is the correct object of the preposition to.
248. HSPT-L5-010 - Answer D: Walking through the orchard, Lena noticed the sweet smell of ripe peaches.
The introductory phrase logically modifies Lena, the person walking.
249. HSPT-L5-011 - Answer D: The conclusion surprised us.
Conclusion already implies a final result, so the shorter sentence avoids redundancy.
250. HSPT-L5-012 - Answer D: The role requires answering calls, keeping accurate records, and scheduling visits.
All three items use parallel gerund phrases.
251. HSPT-L5-013 - Answer B: Sentence 3
The classroom's windows and floor do not contribute to planning or operating the weather station.
252. HSPT-L5-014 - Answer B: Building a useful school weather station requires thoughtful equipment, placement, and recordkeeping.
This sentence previews the passage's central steps and purpose.
253. HSPT-L5-015 - Answer A: After sentence 6
The sentence directly explains why the fixed observation schedule in sentence 6 is important.
254. HSPT-L5-016 - Answer C: Nora packed a flashlight, a map, and a first-aid kit.
Commas correctly separate the three items in the series.
255. HSPT-L5-017 - Answer A: The rain stopped; however, the field remained muddy.
A semicolon joins the independent clauses, and a comma follows the conjunctive adverb however.
256. HSPT-L5-018 - Answer C: Mina said, "The workshop begins at nine."
A comma introduces the quotation, and the period belongs inside the closing quotation mark.
257. HSPT-L5-019 - Answer C: We drove west on Franklin Avenue toward Lake Monroe.
A general direction is lowercase, while complete street and lake names are capitalized.
258. HSPT-L5-020 - Answer B: On Monday, Aunt Rosa will meet Mayor Chen at City Hall.
The day, family title used as a name, official title before a name, and building name are capitalized.
259. HSPT-L5-021 - Answer A: embarrass
Embarrass is spelled with two r's and two s's.
260. HSPT-L5-022 - Answer C: Her response was immediately understood.
Immediately is spelled with two m's at the beginning.
261. HSPT-L5-023 - Answer D: When Priya reached the station, the train had already departed.
Past perfect had departed correctly identifies the action completed before Priya reached the station.
262. HSPT-L5-024 - Answer C: This path is narrower than the other one.
Narrower is the correct comparative form for comparing two paths.

263. HSPT-L5-025 - Answer B: The guide opened the gate when the group arrived.
This choice contains an independent clause and a complete thought.
264. HSPT-L5-026 - Answer D: secured
Secured precisely means fastened or made the loose connection safe.
265. HSPT-L5-027 - Answer C: We returned to the starting point.
Returned already means went back, so adding back is redundant.
266. HSPT-L5-028 - Answer A: Sentence 4
The migration distance of wetland birds does not advance the explanation of the repair project.
267. HSPT-L5-029 - Answer B: Repairing the boardwalk required careful work that protected the wetland while restoring safe access.
The passage emphasizes both the repair steps and protection of the site.
268. HSPT-L5-030 - Answer A: After sentence 3
The sentence explains the benefit of marking and measuring the damaged sections in sentence 3.
269. HSPT-L5-031 - Answer A: If the temperature drops below freezing, cover the young plants.
A comma correctly separates the introductory dependent clause from the command.
270. HSPT-L5-032 - Answer B: The children's projects filled the hallway.
Children is an irregular plural, so its possessive form is children's.
271. HSPT-L5-033 - Answer A: My cousin Elena studies French and world history.
The name and language are capitalized; the family relationship and general course subject are lowercase.
272. HSPT-L5-034 - Answer B: We read "A Wrinkle in Time" during English class.
The important words in the title and the language course name are capitalized correctly.
273. HSPT-L5-035 - Answer B: questionnaire
Questionnaire contains double n and ends with -naire.
274. HSPT-L5-036 - Answer D: independent
Independent is the standard spelling.
275. HSPT-L5-037 - Answer A: The experiment produced an unexpected occurrence.
Occurrence is spelled with two c's and two r's.
276. HSPT-L5-038 - Answer D: Neither of the answers seems complete.
Neither is singular here and takes the singular verb seems.
277. HSPT-L5-039 - Answer D: Jordan and he completed the model.
He is the correct subject pronoun in the compound subject.
278. HSPT-L5-040 - Answer C: The choir sang beautifully during the ceremony.
The adverb beautifully correctly modifies the verb sang.
279. HSPT-L5-041 - Answer D: The hawk soared above the field.
Soared precisely describes smooth flight high in the air.
280. HSPT-L5-042 - Answer A: The students measured the boards, marked the cuts, and assembled the frame.
All three actions use parallel past-tense verbs.
281. HSPT-L5-043 - Answer A: The coach told Maya, "You will lead the warm-up."
Direct quotation removes ambiguity about who will lead.
282. HSPT-L5-044 - Answer B: A repair table extends the life of useful objects while teaching practical skills.
This sentence introduces both the waste-reduction and educational purposes developed in the passage.
283. HSPT-L5-045 - Answer B: Sentence 3
Sentence 3 explicitly says volunteers explain repairs so owners can learn skills.
284. HSPT-L5-046 - Answer B: After the final bell rang, the hallway became quiet.
A comma follows the introductory dependent clause.
285. HSPT-L5-047 - Answer C: The path was steep; nevertheless, the group continued.
A semicolon joins the clauses, and a comma follows nevertheless.
286. HSPT-L5-048 - Answer C: In December, Senator Ortiz spoke at Northview Middle School.
The month, title before a name, surname, and complete school name are capitalized.
287. HSPT-L5-049 - Answer A: The Mississippi River flows south toward the Gulf of Mexico.
The complete geographic names are capitalized, while the general direction south is lowercase.

288. HSPT-L5-050 - Answer D: perseverance
Perseverance is the standard spelling.
289. HSPT-L5-051 - Answer C: liaison
Liaison is spelled l-i-a-i-s-o-n.
290. HSPT-L5-052 - Answer A: The laboratory purchased new equipment.
Equipment is spelled without a t after p.
291. HSPT-L5-053 - Answer C: Last week, the class visited the planetarium.
The past-time expression last week requires the past-tense verb visited.
292. HSPT-L5-054 - Answer D: The trail was flooded, so the hikers chose another route.
A comma before the coordinating conjunction so correctly joins the two independent clauses.
293. HSPT-L5-055 - Answer C: The hikers admired the mountain covered with fresh snow.
The phrase covered with fresh snow is placed directly after mountain, the noun it modifies.
294. HSPT-L5-056 - Answer D: confirm
Confirm means establish that a conclusion is correct or well supported.
295. HSPT-L5-057 - Answer C: The two teams cooperated on the project.
Cooperated already means worked together, so together is unnecessary.
296. HSPT-L5-058 - Answer A: The weather may affect the timing of the event.
Affect is the verb meaning to influence.
297. HSPT-L5-059 - Answer B: Seeds from the same species may have been grown under different conditions.
Location information allows later comparisons among plants grown in different places.
298. HSPT-L5-060 - Answer B: After sentence 2
The sentence explains why the identifying details listed in sentence 2 are necessary.